Liver Transplantation

When to Refer and What's New

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Slides (PDF)

Illustrations generated with Google Gemini



Disclosures

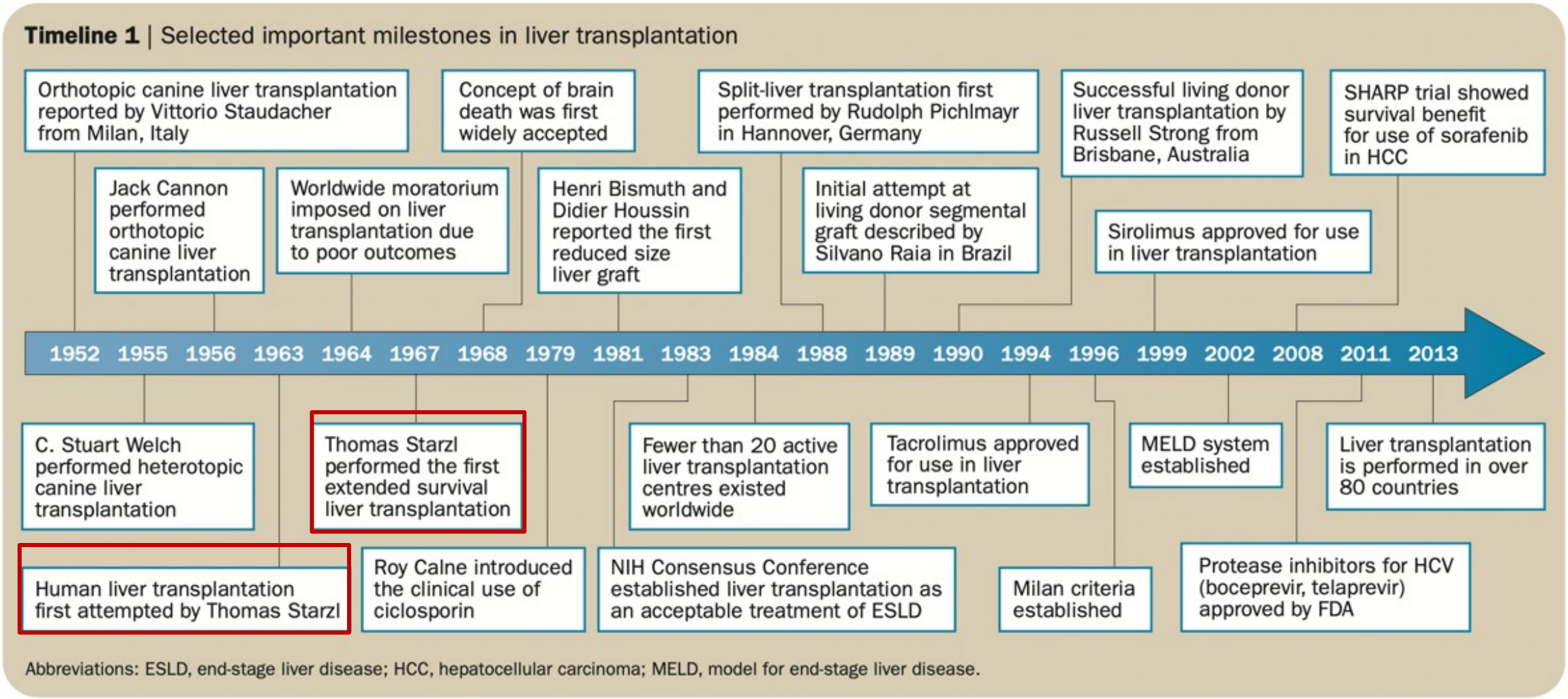
No relevant financial relationships to disclose.

Objectives

- Recognize who to refer, and who needs a call the same day
- See who can be a transplant candidate now
- Walk through what happens after the referral
- Understand where donor livers come from now
- Care for a transplant recipient on your service



How did liver transplant start?



Timeline 1 from Zarrinpar A, Busuttil RW. Nat Rev Gastroenterol Hepatol 2013;10(7):434-40 [PMID: 23752825]

Who was Thomas Starzl?


The New York Times

Dr. Thomas E. Starzl, Pioneering Liver Surgeon, Dies at 90

- 1963, Denver: first human liver transplant attempt
- 1967: first patient to live more than a year after one
- 1980: cyclosporine with steroids; survival finally good enough to offer it widely
- 1989, Pittsburgh: tacrolimus
- Died March 4, 2017, at 90

The New York Times, March 5, 2017. Starzl TE et al. Homotransplantation of the liver in humans. Surg Gynecol Obstet 1963;117:659-76 [PMID: 14100514]

How did the first liver transplants go?

SURGERY
Gynecology & Obstetrics

DECEMBER 1963
VOLUME 117
NUMBER 6

HOMOTRANSPLANTATION OF THE LIVER IN HUMANS

T. E. STARZL, M.D., F.A.C.S., T. L. MARCHIORO, M.D., K. N. VON KAULLA, M.D., G. HERMANN, M.D., R. S. BRITTAIN, M.D., and W. R. WADDELL, M.D., F.A.C.S., Denver, Colorado

- Patient 1, 1 March 1963: a 3-year-old with biliary atresia, died of bleeding on the operating table
- Patients 2 and 3 died 22 and 7½ days later, of pulmonary emboli

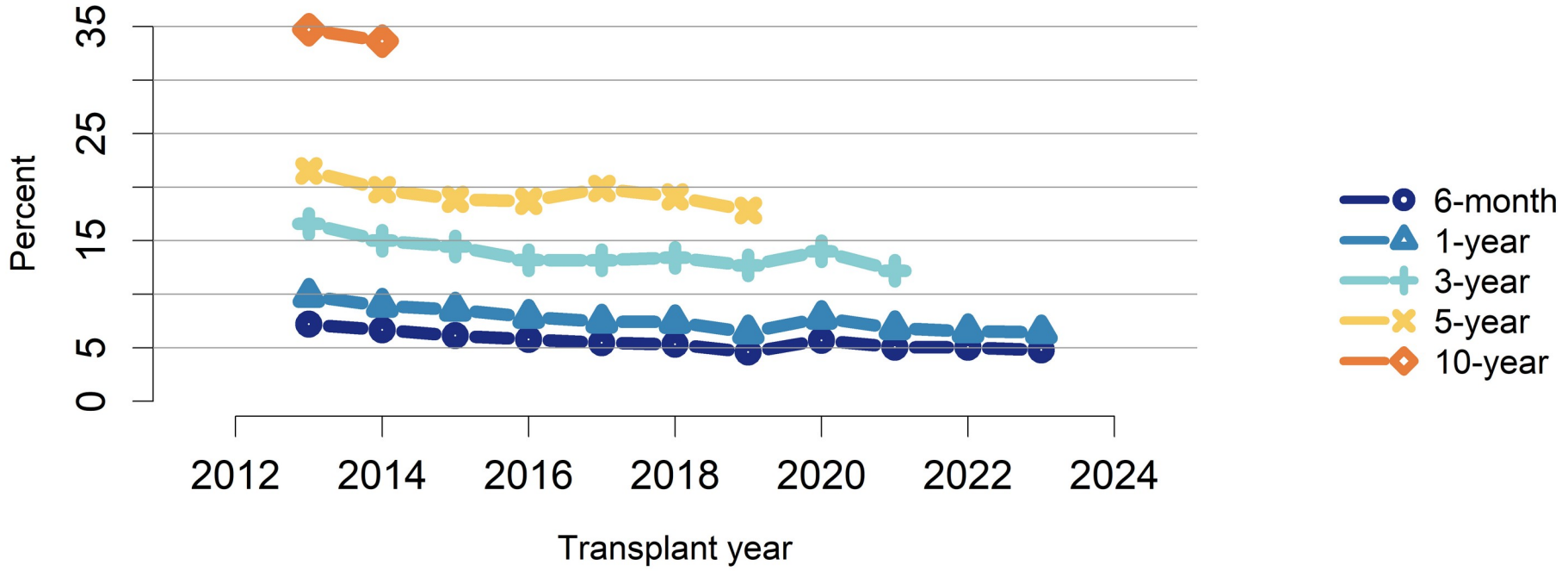
The first attempt resulted in failure at the operating table. The course of the second 2 patients establishes the feasibility of such an operation in humans, despite the fact that death occurred 22 and 7½ days after transplantation from pulmonary emboli.

Recipient patients. Patient 1 was a 3 year old white male with congenital biliary atresia (Fig. 1A). Physical development had been retarded, preoperative weight being 20 pounds. His general condition was poor, with hepatosplenomegaly, jaundice, and ascites. Total bilirubin was 20.7 milligrams per cent with a conjugated fraction of 16.7 milligrams per cent. Alkaline phosphatase was 12.8 Bodansky units. Serum glutamic-oxalacetic acid transaminase (SGOT) was 160 SF units. The hospital course prior to hepatic homotransplantation on 1 March 1963 was uneventful.

Starzl TE et al. Surg Gynecol Obstet 1963;117:659-76 [PMID: 14100514]. Scan: University of Pittsburgh D-Scholarship

How do patients do after a liver transplant?

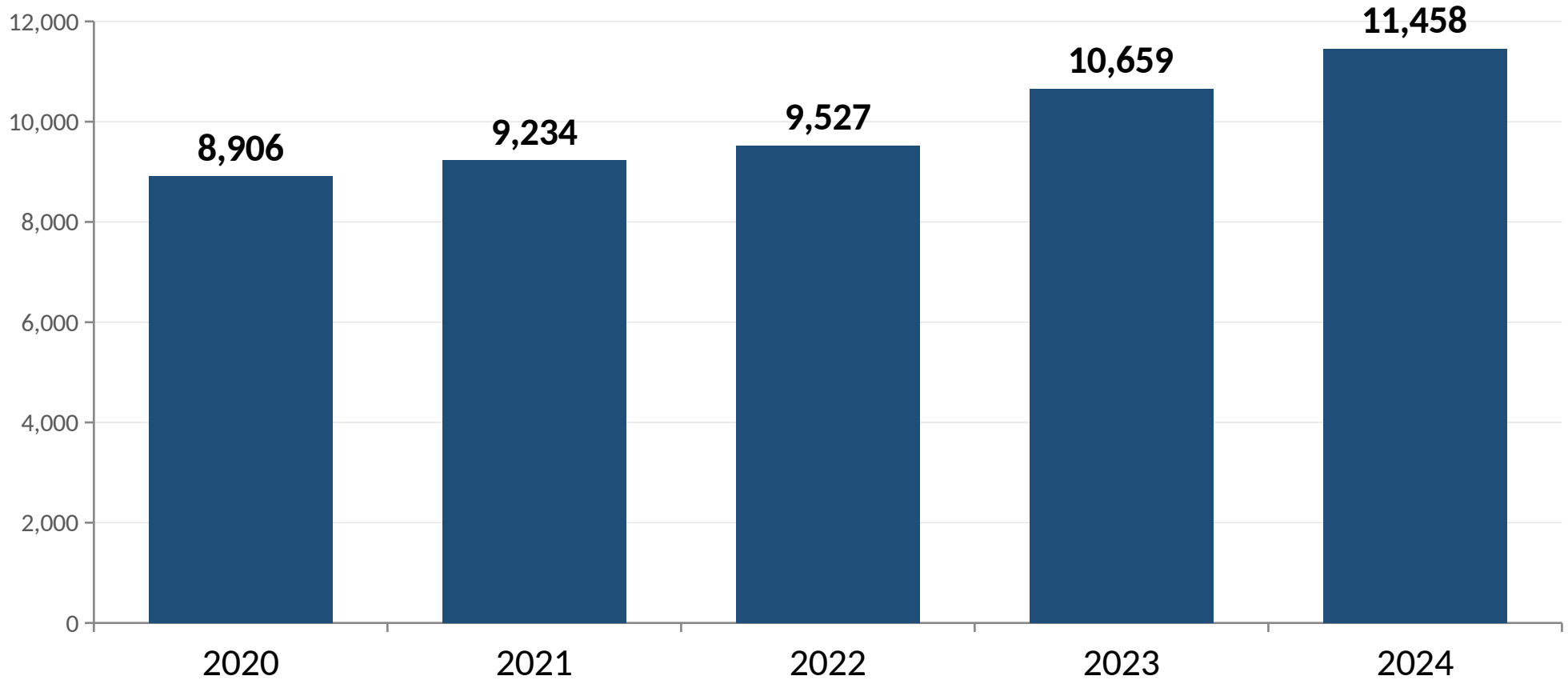
Figure LI 59: Percentages of patient deaths among adult liver transplant recipients



OPTN/SRTR 2024 Annual Data Report

OPTN/SRTR 2024 Annual Data Report, Figure LI 59. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

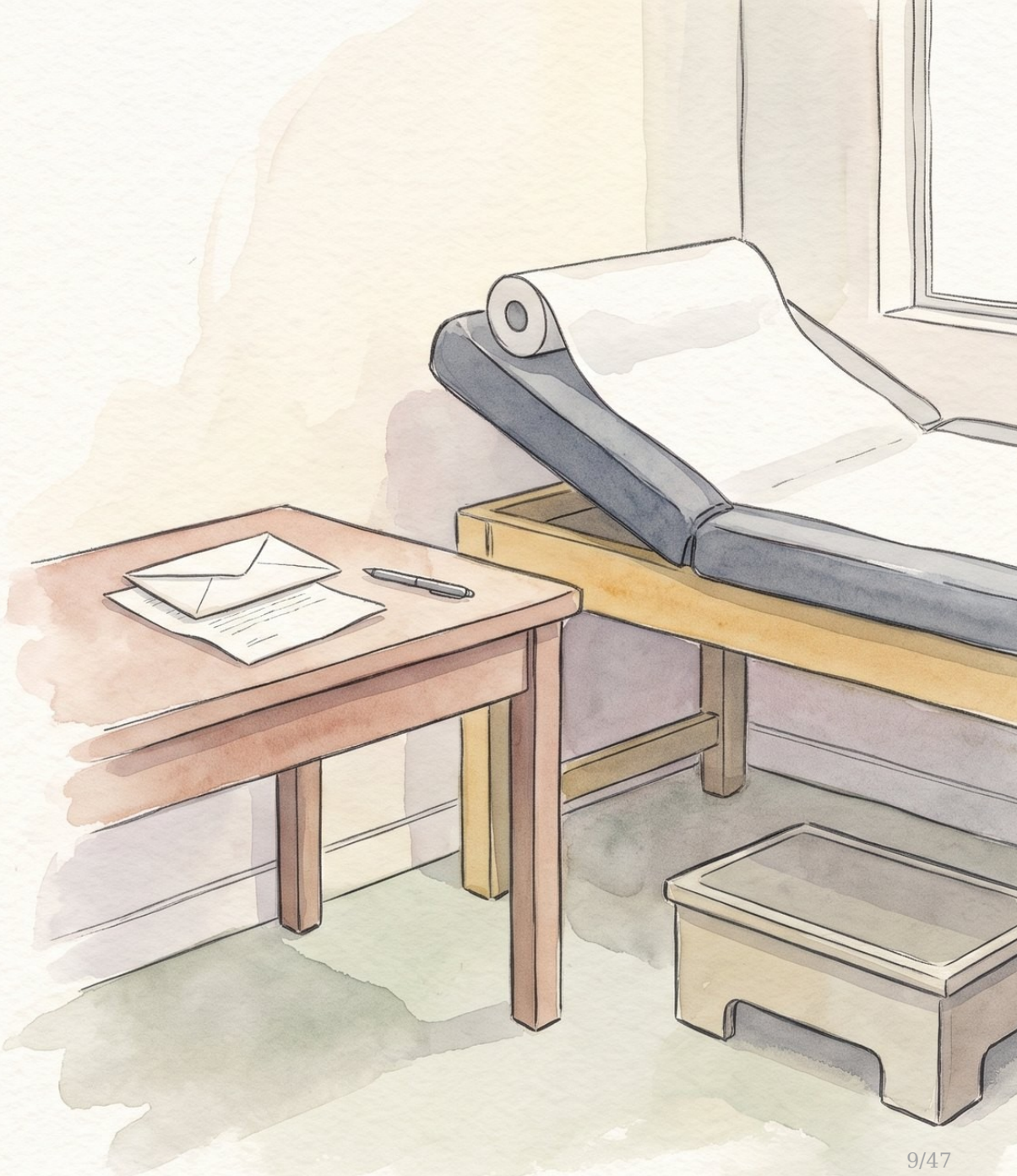
How much has liver transplant grown?



11,458 in 2024 (10,886 adults), 70% more than in 2014

OPTN/SRTR Annual Data Reports, Liver, 2020-2024 [PMIDs: 35266621, 37132348, 38431359, 39947804, 42276697]

When to refer



Where did MELD come from?

Comparative Study > Hepatology. 2000 Apr;31(4):864-71. doi: 10.1053/he.2000.5852.

A model to predict poor survival in patients undergoing transjugular intrahepatic portosystemic shunts

M Malinchoc [1], P S Kamath, F D Gordon, C J Peine, J Rank, P C ter Borg

Review > Hepatology. 2001 Feb;33(2):464-70. doi: 10.1053/jhep.2001.22172.

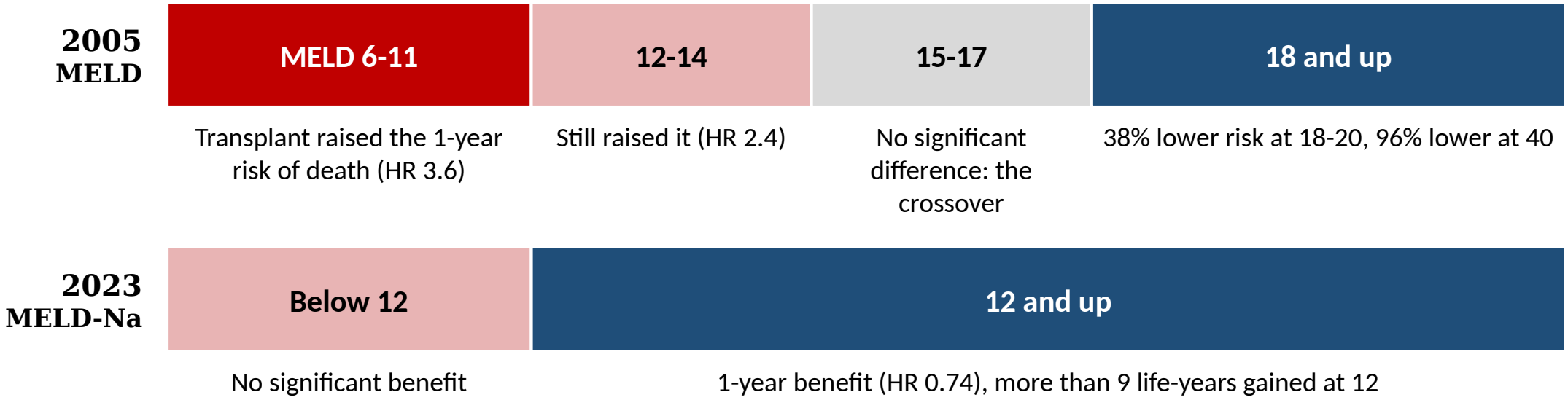
A model to predict survival in patients with end-stage liver disease

P S Kamath [1], R H Wiesner, M Malinchoc, W Kremers, T M Therneau, C L Kosberg, G D'Amico, E R Dickson, W R Kim

- 2000: built at Mayo to predict death after an elective TIPS
- 2001: worked just as well across end-stage liver disease
- 2002: became the basis for US liver allocation

Malinchoc M, Kamath PS, Gordon FD, et al. Hepatology 2000;31(4):864-71 [PMID: 10733541]. Kamath PS et al. Hepatology 2001;33(2):464-70 [PMID: 11172350]

Why did MELD 15 become the rule?



Neither number has been set for MELD 3.0. Referral is a separate question: refer after any decompensating event, whatever the MELD. Without one, 15 is still reasonable.

Merion RM et al. Am J Transplant 2005;5(2):307-13 [PMID: 15643990]. Malamon JS et al. Int J Surg 2023;109(9):2714-20 [PMID: 37226874]. Dove L et al. Hepatology 2026, statement 1 [PMID: 41405234]

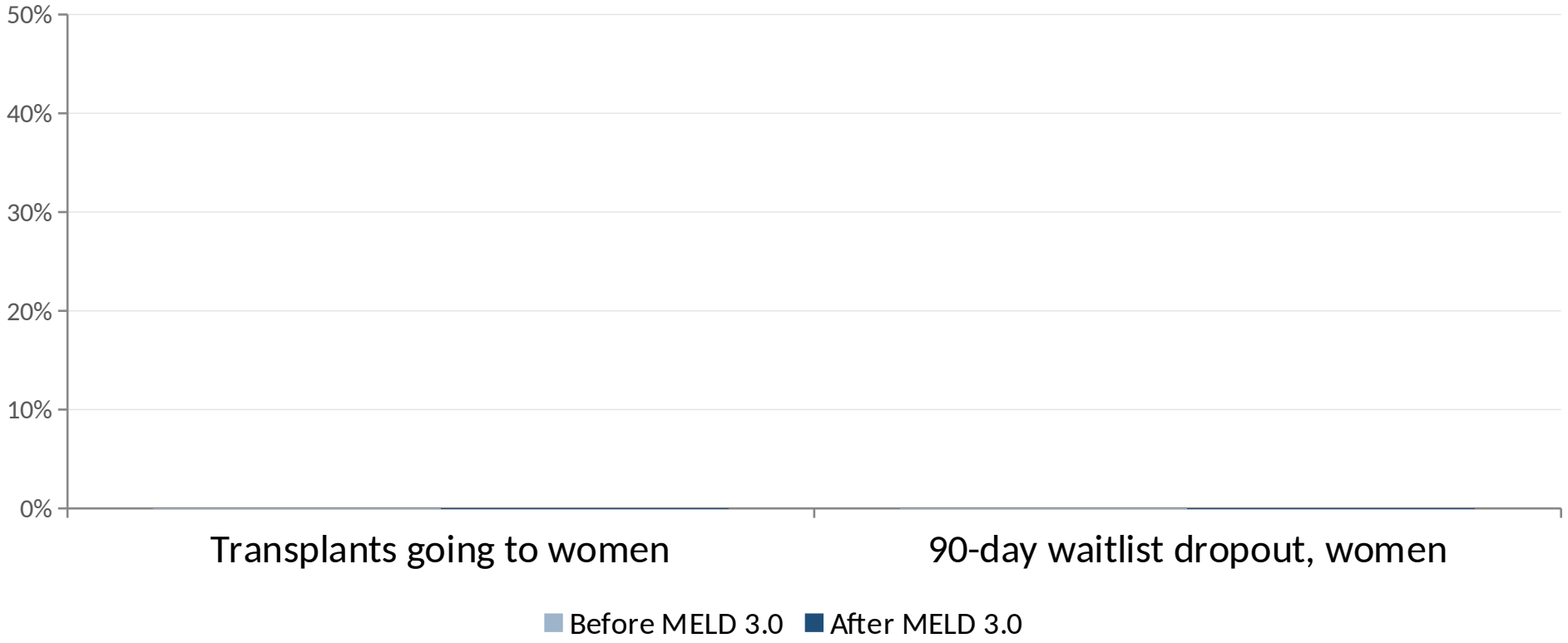
What's new in MELD 3.0?

	MELD-Na	MELD 3.0
Bilirubin	✓	✓
INR	✓	✓
Creatinine	✓ (cap 4.0)	✓ (cap 3.0)
Sodium	✓	✓
Albumin		✓
Female sex		✓ **(+1.33)**
Interaction terms	MELD × Na	Bili × Na, albumin × Cr

- Used for allocation since July 2023
- 90-day mortality: C-statistic 0.869 vs 0.862
- In simulation, at least 20 fewer waitlist deaths a year

Kim WR, Mannalithara A, Heimbach JK, et al. Gastroenterology 2021;161(6):1887-1895.e4 [PMID: 34481845]

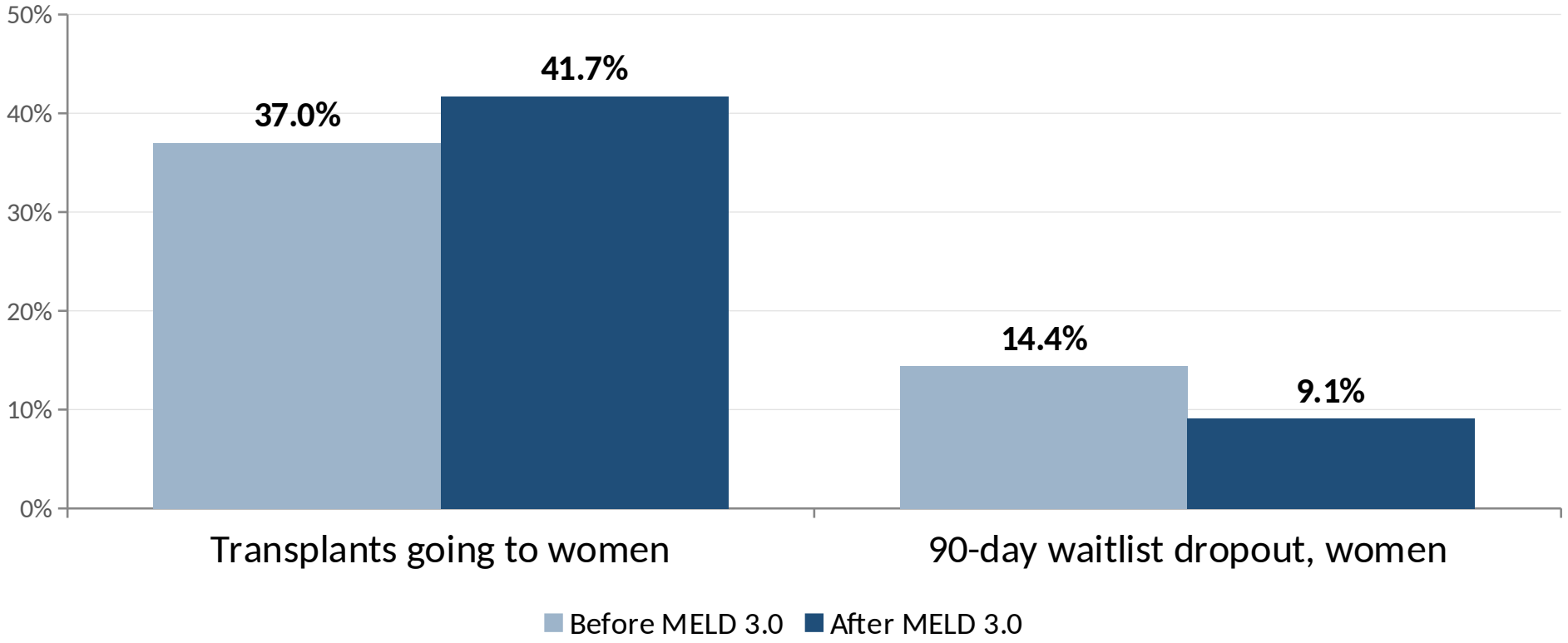
Did MELD 3.0 help women?



Conference analysis, 12 months before vs 12 months after July 2023. Dropout: died or too sick for transplant

Kwong AJ, The Liver Meeting 2024, abstract 5002 (conference data). Tanaka T et al. Hepatology 2026;84(1):106-16 [PMID: 41100798]. Di Napoli M et al. Front Transplant 2026 [PMID: 42266471]. SRTR 2024 ADR [PMID: 42276697]

Did MELD 3.0 help women?

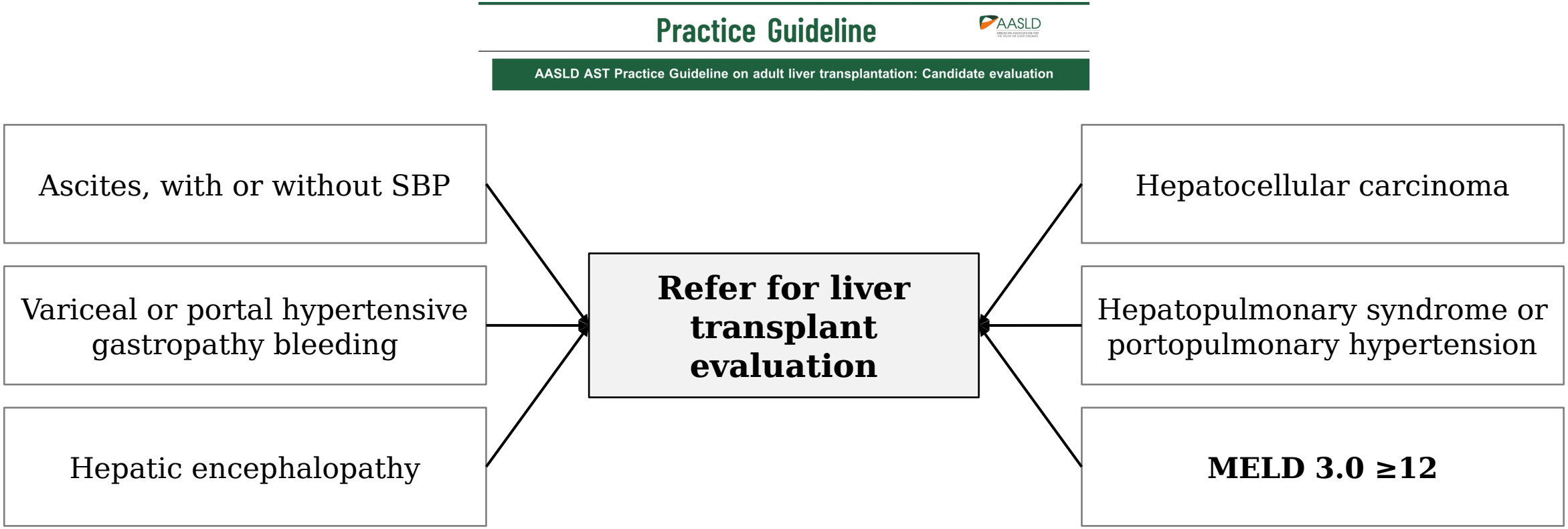


Conference analysis, 12 months before vs 12 months after July 2023. Dropout: died or too sick for transplant

The gain was largest right after the change and narrowed over time. Waitlisted women still die more often: 15.0 vs 12.4 per 100 patient-years in 2024.

Kwong AJ, The Liver Meeting 2024, abstract 5002 (conference data). Tanaka T et al. Hepatology 2026;84(1):106-16 [PMID: 41100798]. Di Napoli M et al. Front Transplant 2026 [PMID: 42266471]. SRTR 2024 ADR [PMID: 42276697]

When should a patient with cirrhosis be referred?



The 2025 guideline: refer after any decompensating event. Without one, the MELD threshold is unclear: benefit may start at 12 in waitlisted patients, and most of the writing group still sees 15 as reasonable. “MELD score alone should not be a barrier to referral.”

Figure 1 redrawn from Dove L et al. AASLD/AST candidate evaluation guideline. Hepatology 2026;83(6):1609-45 [PMID: 41405234], statement 1

Who needs a call the same day?

Acute liver failure
INR ≥1.5 and any encephalopathy, no cirrhosis

Severe acute liver injury
Jaundice and a rising INR, still awake

ACLF
Cirrhosis with new organ failure

Call transplant hepatology



ALF: call “in the first few hours after presentation, and prior to meeting ALF diagnostic criteria.” Insurance, substance use or infection shouldn't hold up the call.

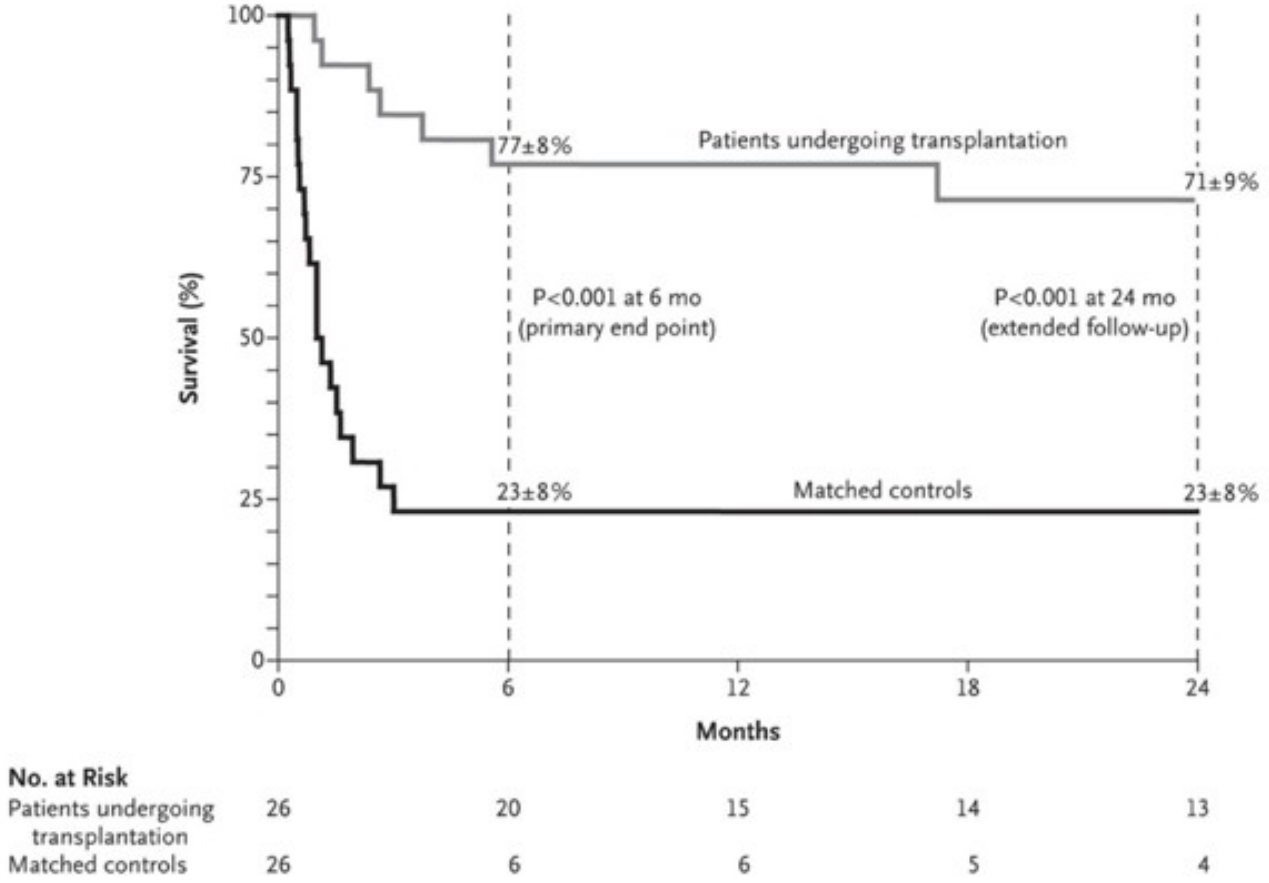
ACLF: consider referral early, while the patient can still travel.

Dove L et al. AASLD/AST candidate evaluation guideline. Hepatology 2026;83(6):1609-45 [PMID: 41405234], statements 2-4. ACG ALF guideline 2023 [PMID: 37377263]

Who can get a liver now



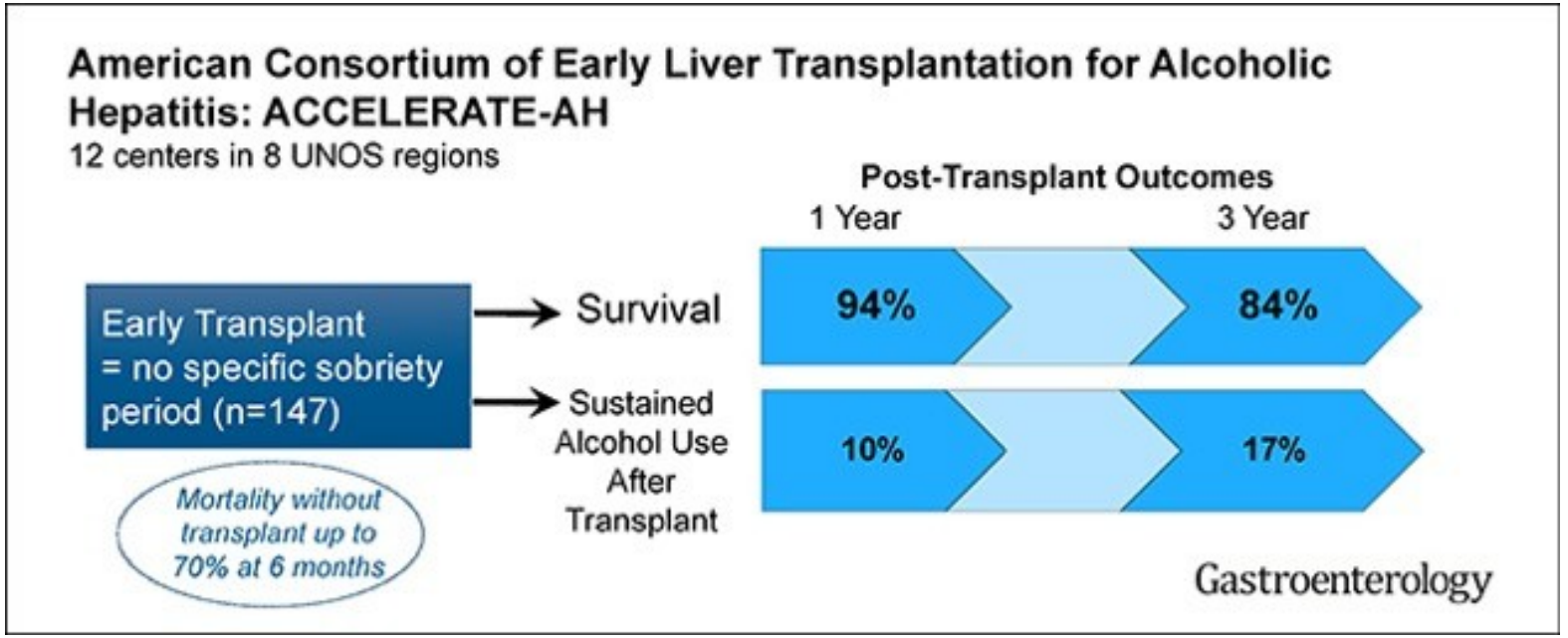
Can a patient who drank last month get a liver?



- **France and Belgium, not the US**
- First episode of severe AH, not responding to steroids, strict medical and psychosocial selection
- 26 patients, fewer than 2% of severe AH admissions
- 6-month survival 77% vs 23% in matched controls

Mathurin P, Moreno C, Samuel D, et al. N Engl J Med 2011;365(19):1790-800 [PMID: 22070476]

Did it hold up in the US?

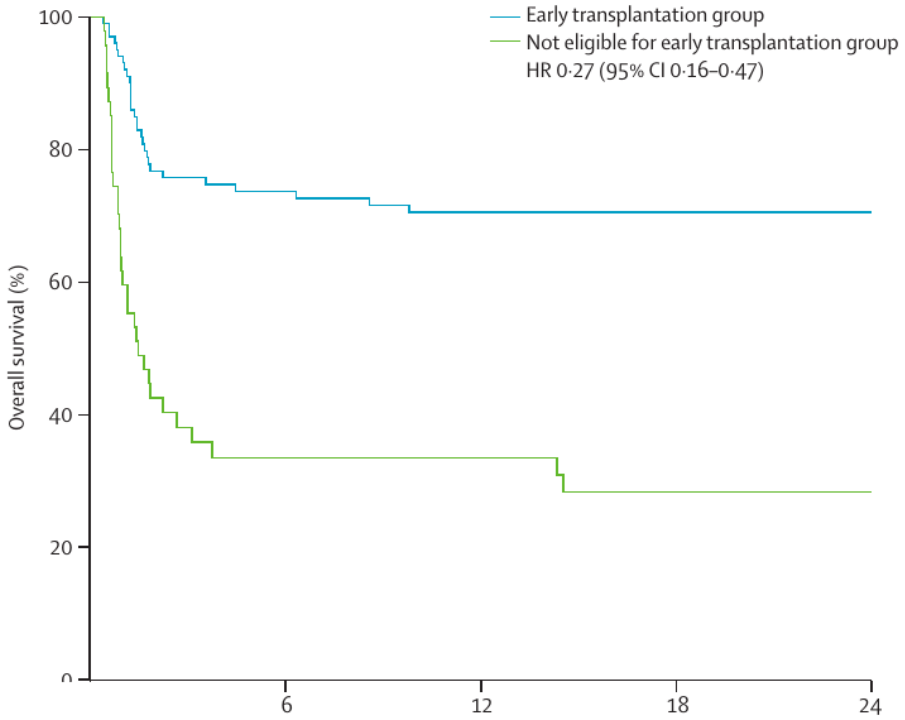


- ACCELERATE-AH, 147 patients: sustained alcohol use carried a higher risk of death (HR 4.59)
- 2025 analysis, 347 patients alive at discharge: 5-year survival 83.1%

Lee BP et al. Gastroenterology 2018;155(2):422-430.e1 [PMID: 29655837]. Dukewich M et al. Am J Gastroenterol 2025;120(4):827-36 [PMID: 38994850]

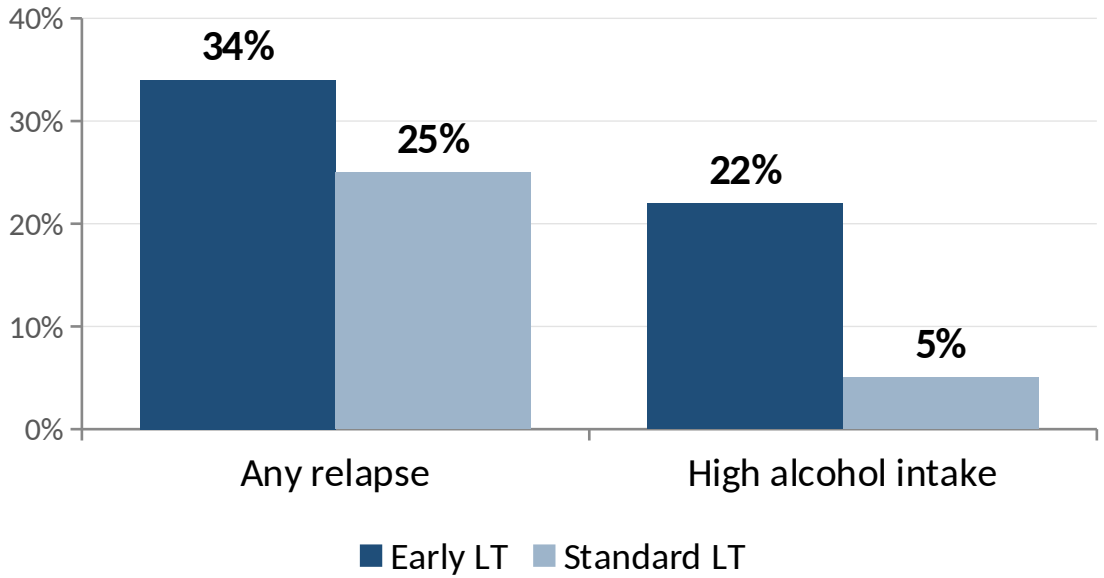
What about relapse?

Early liver transplantation for severe alcohol-related hepatitis not responding to medical treatment: a prospective controlled study



Early-LT strategy vs patients not eligible: 2-year survival 70.6% vs 28.3%

Relapse in transplanted patients: early vs standard LT



Non-inferiority for relapse not shown

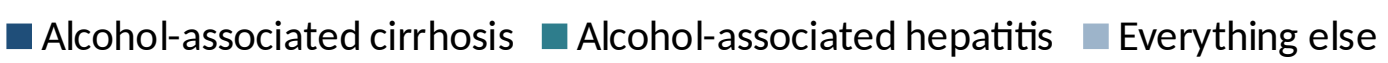
Louvet A, Labreuche J, Moreno C, et al. Lancet Gastroenterol Hepatol 2022;7(5):416-25 [PMID: 35202597]

What do the guidelines say now?

Guideline	Early transplant for alcohol-associated hepatitis
EASL 2024	Selected non-responders with favorable sobriety predictors (strong)
ACG 2024	Carefully selected non-responders at low risk of relapse
AASLD/AST 2025	Refer ALD early; abstinence duration alone doesn't exclude

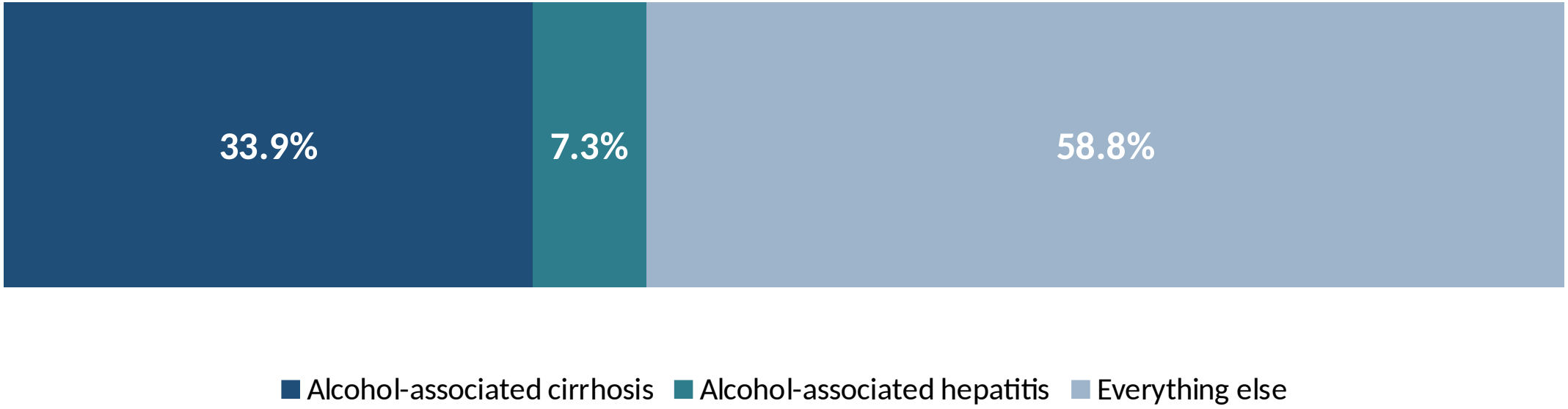
EASL CPG on liver transplantation. J Hepatol 2024;81(6):1040-86, p.1044 [PMID: 39487043]. Jophlin LL et al. Am J Gastroenterol 2024;119(1):30-54 [PMID: 38174913]. Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234], statements 37-38

What do US adults get a liver for?



OPTN/SRTR 2024 Annual Data Report, Table LI 7. Kwong AJ et al. OPTN/SRTR 2024 Annual Data Report: Liver. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

What do US adults get a liver for?

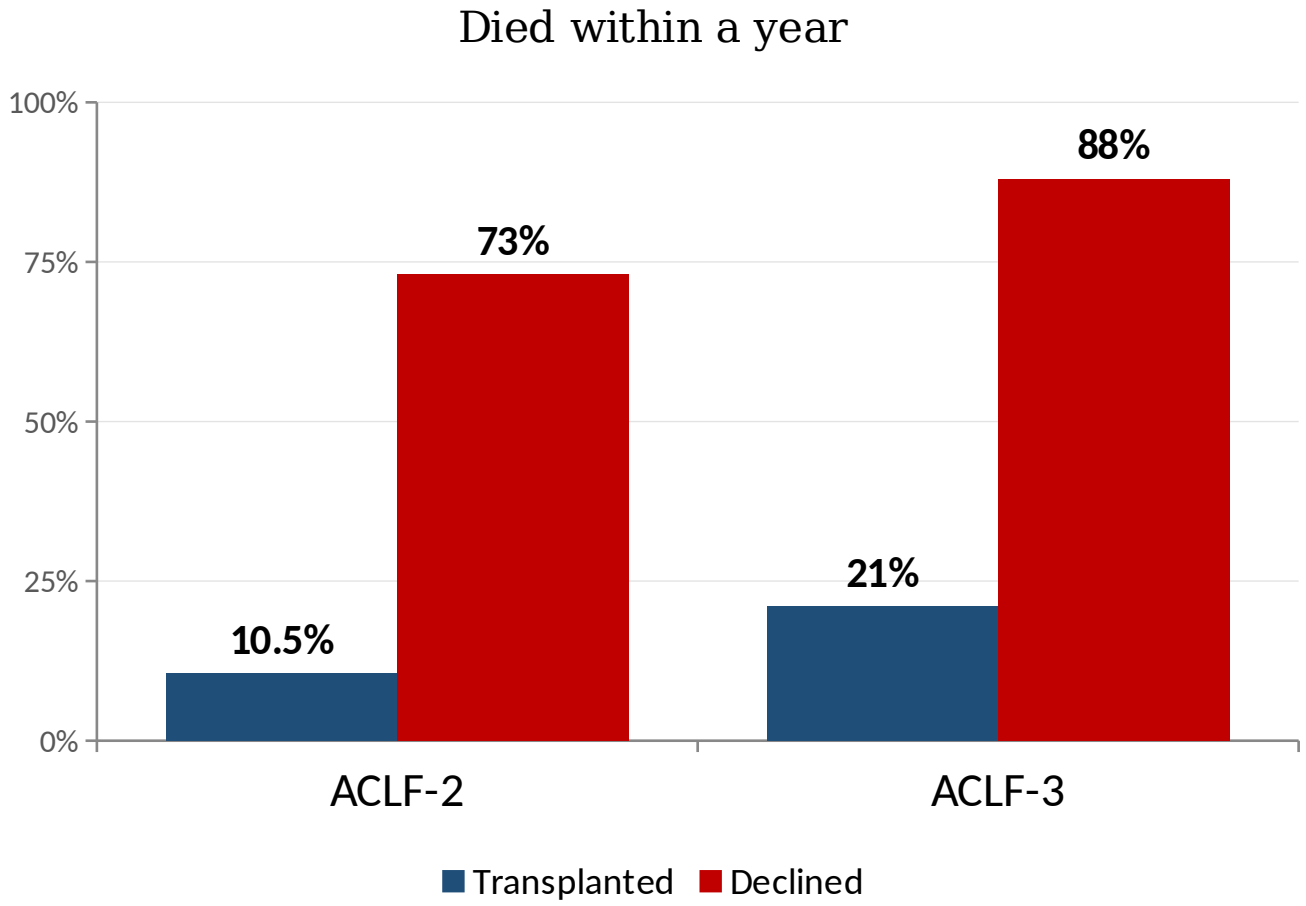


Alcohol-associated liver disease: 41.1% of US adult recipients, the top diagnosis

Parts are rounded, so they add to 41.2%

OPTN/SRTR 2024 Annual Data Report, Table LI 7. Kwong AJ et al. OPTN/SRTR 2024 Annual Data Report: Liver. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

Can patients with ACLF get a liver?



Transplanted: case-fatality, no retransplant. Declined: 1-year mortality

- CHANCE: 612 transplanted, 144 declined, 63 centers
- Without ACLF: 9% case-fatality at a year after transplant
- AASLD 2024: selected patients with ACLF may warrant expedited transplant; palliative care can run alongside the evaluation

Gustot T, Bernal W, Fernández J, et al. CHANCE. J Hepatol 2026 [PMID: 42822699]. Karvellas CJ et al. Hepatology 2024;79(6):1463-1502, statements 47, 51 [PMID: 37939273]

What no longer rules a patient out?

On its own	What the 2025 AASLD/AST guideline says
Length of abstinence	Duration alone doesn't exclude
Methadone or buprenorphine	No denial, no taper needed
BMI >40 or <18.5	Screen nutrition and muscle; BMI alone doesn't exclude
ACLF or MELD >40	Doesn't exclude by itself
Older age, with a living donor	Not an exclusion

None of these is a cutoff on its own.

Dove L et al. AASLD/AST candidate evaluation guideline. Hepatology 2026;83(6):1609-45 [PMID: 41405234], statements 2, 38, 41, 43, 48-49, 51

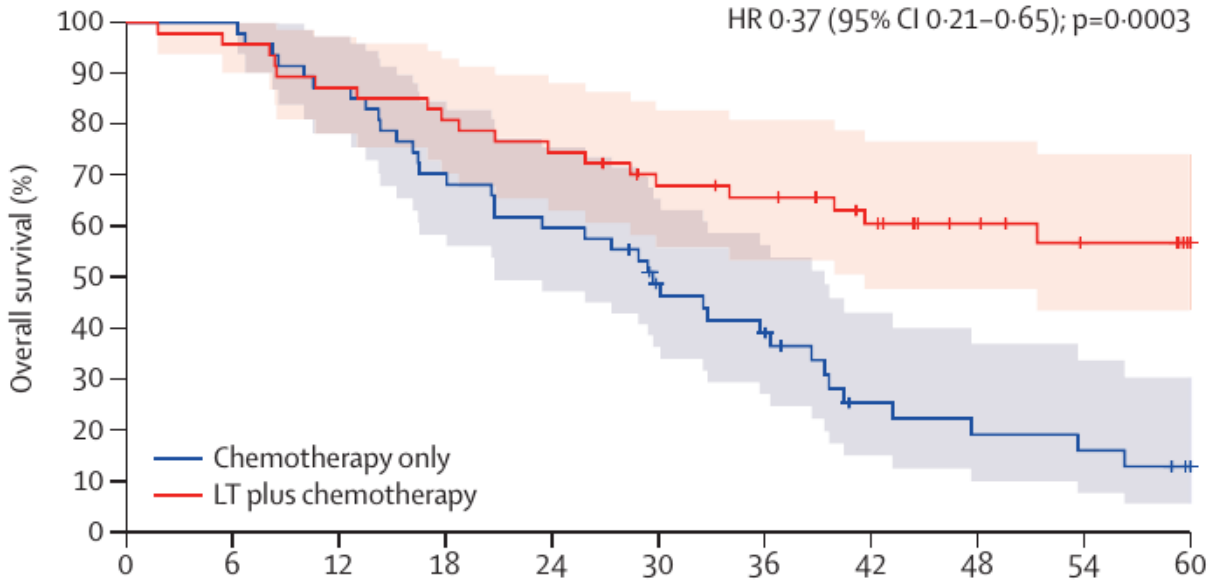
Who gets exception points?

Exception	What's needed
Hepatocellular carcinoma	1 lesion 2-5 cm, or 2-3 lesions each 1-3 cm AFP ≤1000 (if >1000, must fall below 500)
Hepatopulmonary syndrome or portopulmonary hypertension	Confirmed by testing; a standard exception
Colorectal metastases or intrahepatic cholangiocarcinoma	Case by case, through the review board

OPTN Policies 9.5.E, 9.5.G, 9.5.I (effective 10/1/2026); OPTN adult MELD exception transplant oncology review guidance

Can a liver transplant help colorectal metastases?

Liver transplantation plus chemotherapy versus chemotherapy alone in patients with permanently unresectable colorectal liver metastases (TransMet): results from a multicentre, open-label, prospective, randomised controlled trial



- **20 centers in France, Belgium and Italy, not the US**
- 94 randomized: age 18-65, BRAF wild-type, primary resected, permanently unresectable liver-only disease responding to chemotherapy
- 5-year survival (intention to treat): 56.6% with LT plus chemotherapy vs 12.6% with chemotherapy
- 72% of transplanted patients recurred, mostly in the lungs; almost half of recurrences were resected or ablated
- **In the US: 158 transplants at 33 centers, 3-year survival 67%, about half from living donors**

Adam R, Piedvache C, Chiche L, et al. Lancet 2024;404(10458):1107-18 [PMID: 39306468]. Chávez-Villa M et al. Liver Transpl 2026;32(4):508-17 [PMID: 41086433]

From Maine to Boston



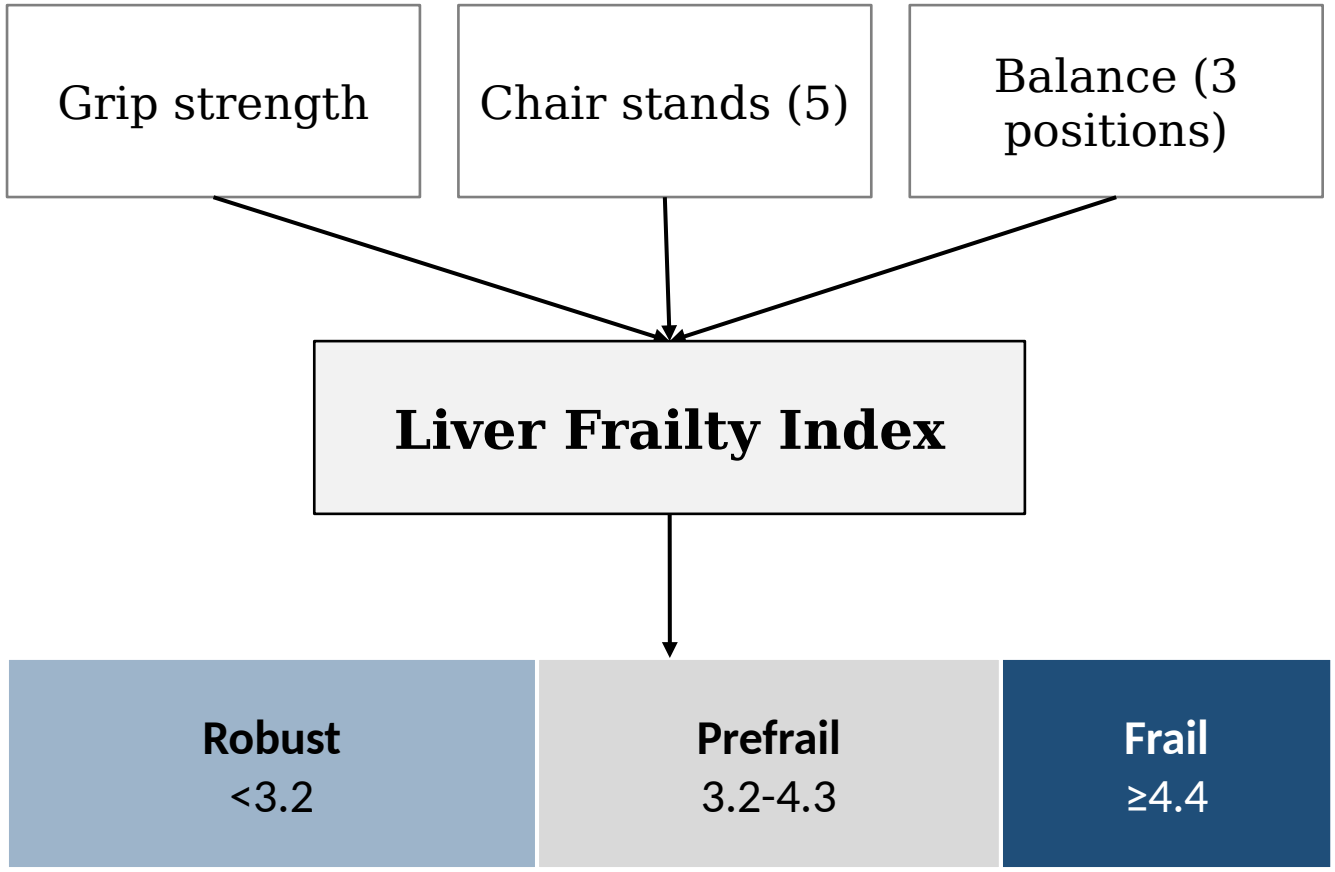
What happens after the referral?



Part of the evaluation	What it involves
Hepatology and surgery	Visits, labs, CT or MRI of the liver
Heart and lungs	ECG and echo, then coronary testing by risk
Social work and addiction	Home support, alcohol and drug use, treatment plan
Nutrition and physical therapy	Muscle, frailty, protein intake
Other	Dental, cancer screening, vaccines, financial counseling

Dove L et al. AASLD/AST candidate evaluation guideline. Hepatology 2026;83(6):1609-45 [PMID: 41405234]

How should frailty change the evaluation?

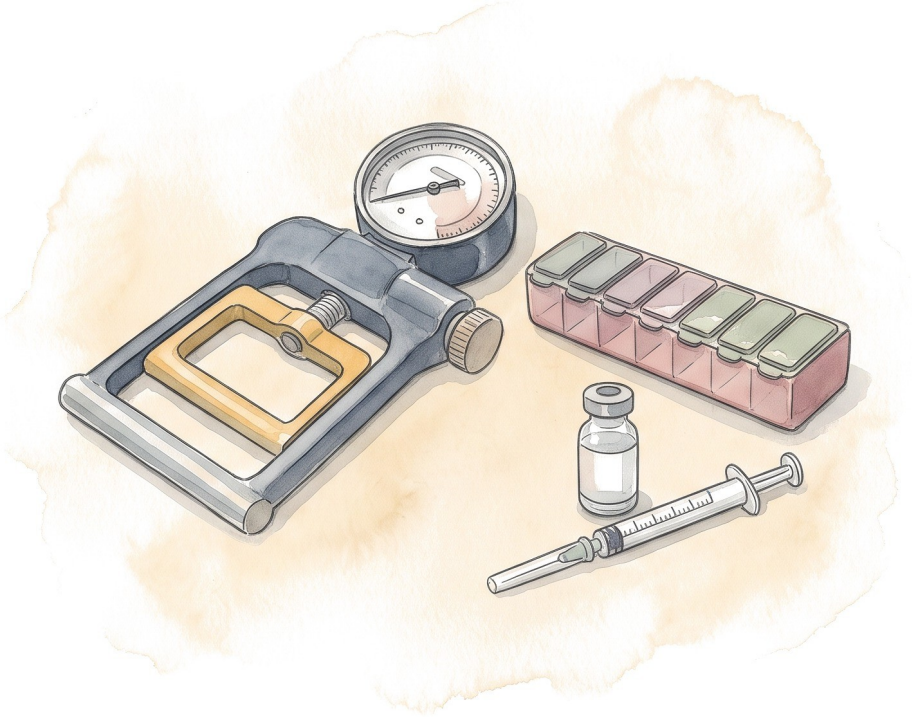


- Use the Liver Frailty Index; it takes 3 minutes
- Frailty raises the risk after transplant, but alone it doesn't rule a patient out
- Frailty also predicts death on the waitlist

Lai JC, Tandon P, Bernal W, et al. Hepatology 2021;74(3):1611-44 [PMID: 34233031]. Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234], statements 33, 44-45

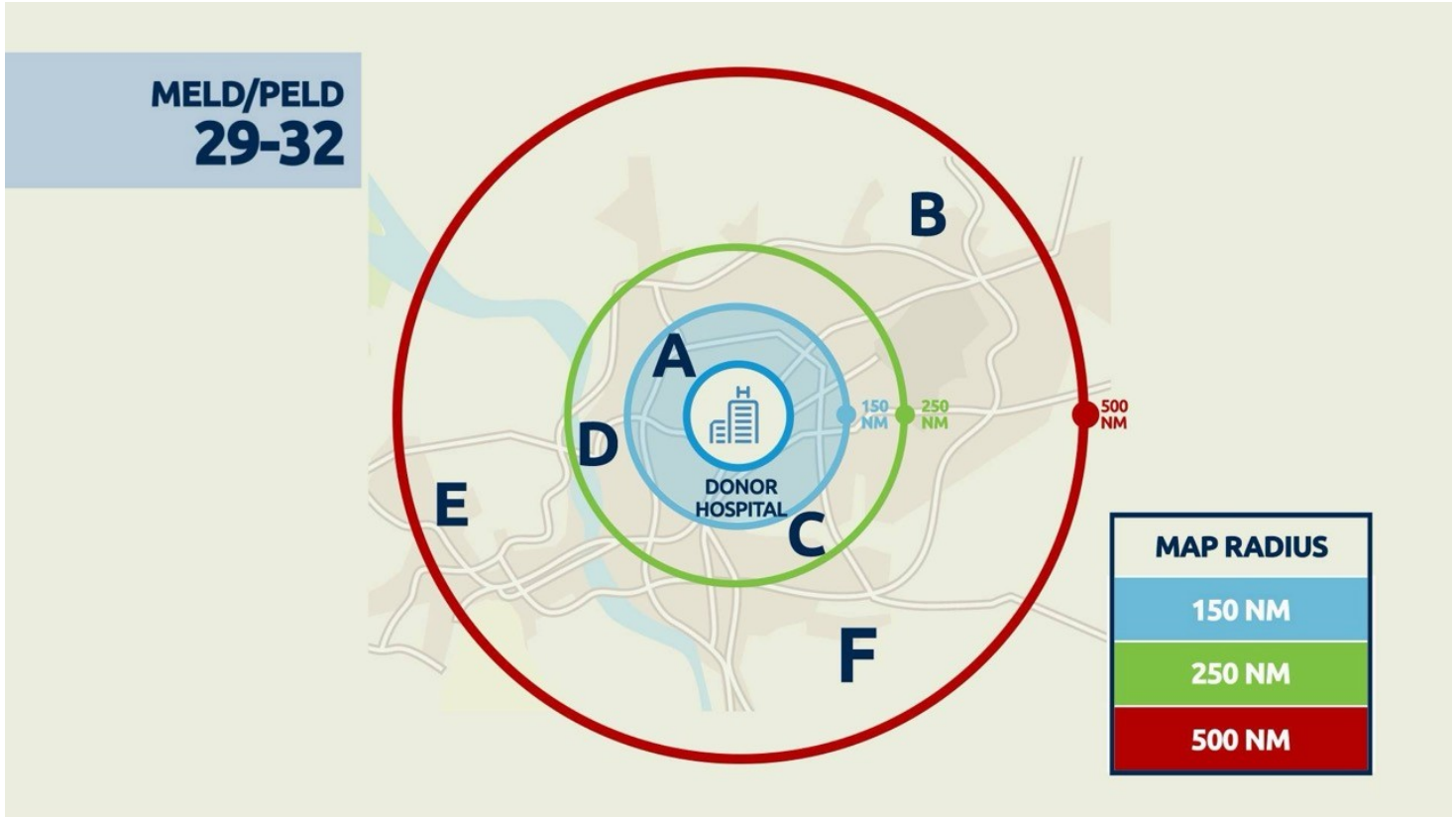
What can you do while the patient waits?

- Treat alcohol use disorder, with medicines and counseling
- Protein 1.2-1.5 g/kg ideal body weight a day, a late-evening snack
- Exercise for frailty
- Vaccines before transplant
- No known HCC: surveillance every 6 months
- Keep the MELD labs current



Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234]. Lai JC et al. Hepatology 2021 [PMID: 34233031]

How does a liver get from a donor to a patient?



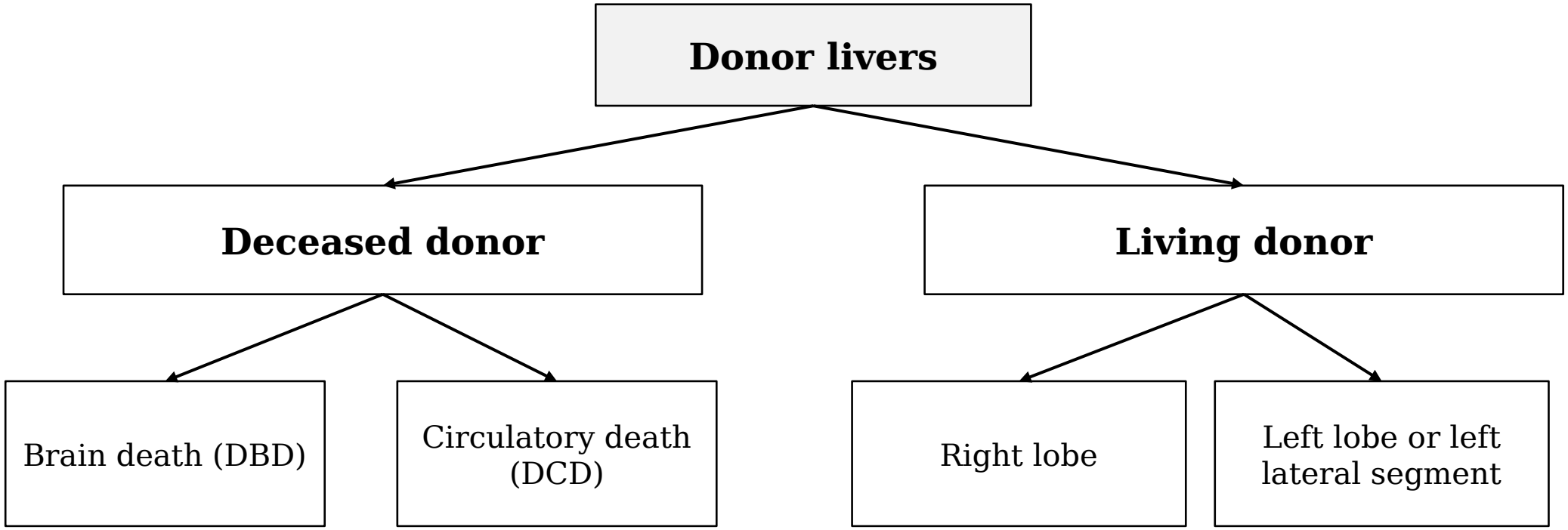
- Circles of 150, 250 and 500 nautical miles around the donor hospital
- Offers follow urgency first, then distance from the donor hospital
- Exception points are set from the median MELD at transplant (MMaT) near the donor hospital

OPTN Policy 9, liver allocation (acuity circles since February 2020)

Where the livers come from



What kinds of donors are there?



Either can go whole or be split

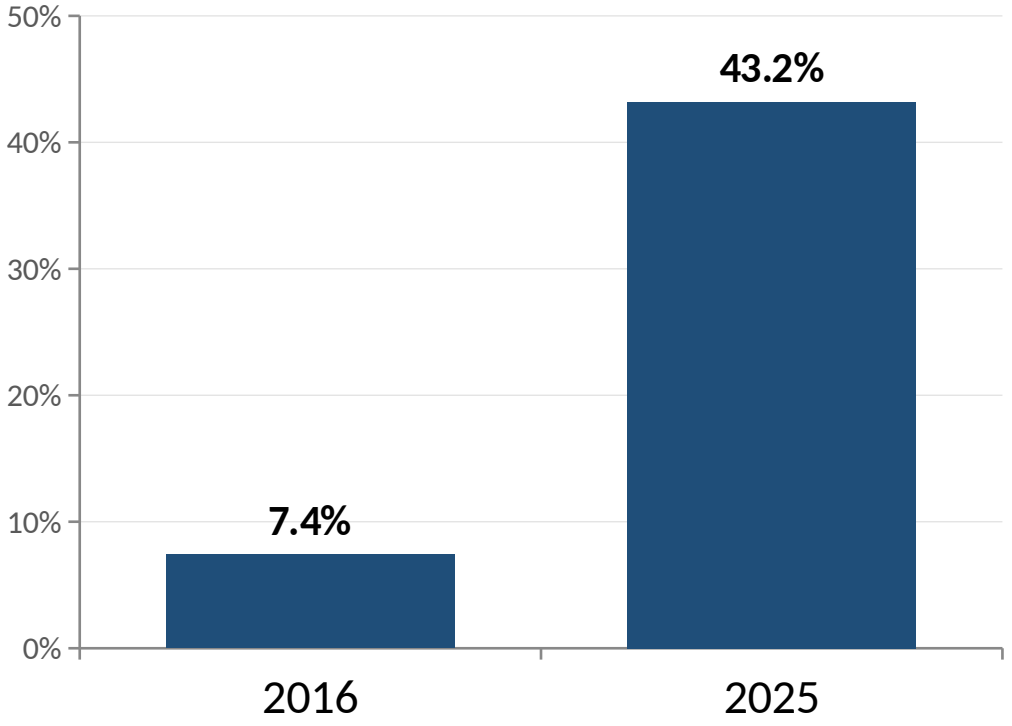
DCD: 43% of US deceased donor liver transplants in 2025

Living donor: 4.5% of US adult liver transplants in 2024

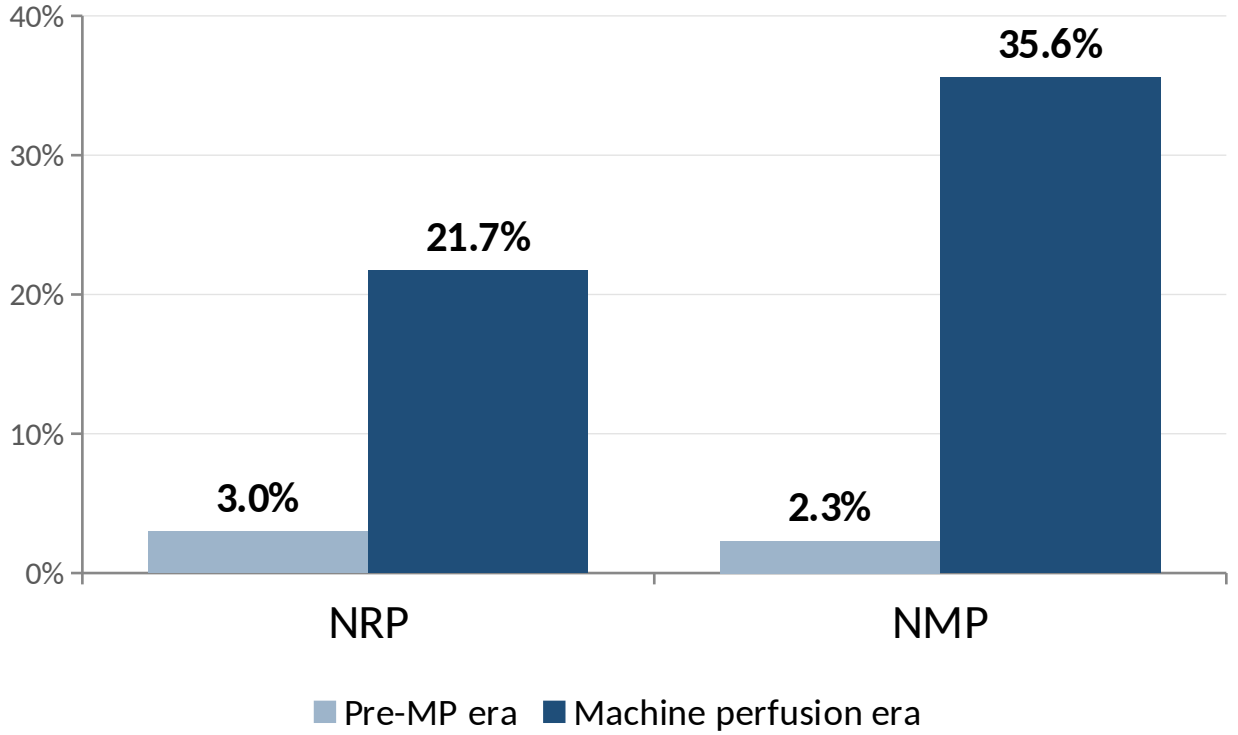
Giorgakis E et al. Transpl Int 2026;39:16689 [PMID: 42294094]. Kwong AJ et al. OPTN/SRTR 2024 Annual Data Report: Liver. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

How much has DCD grown?

DCD share of deceased donor LTs



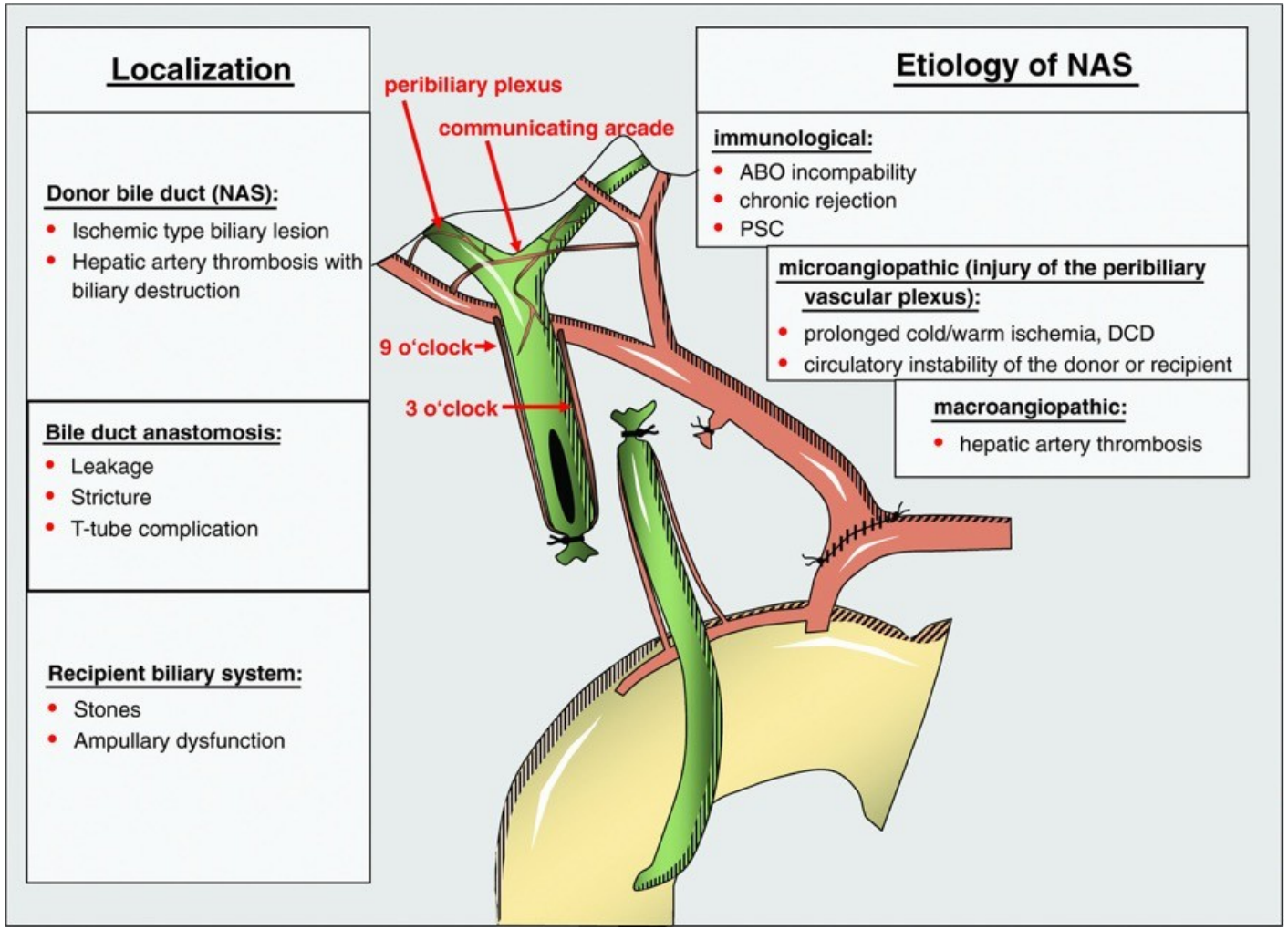
Use of NRP and NMP



Adult DCD recipients: median wait 112 vs 62 days, earlier vs later era

Giorgakis E, Ziogas IA, Moris D, et al. Transpl Int 2026;39:16689 [PMID: 42294094]

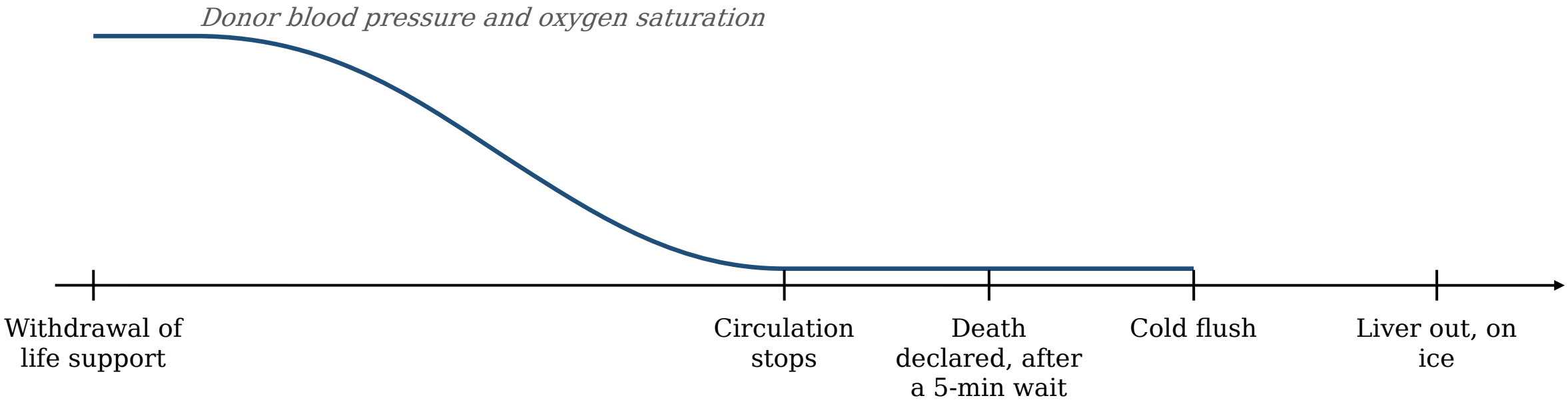
Why did DCD livers worry us?



- After transplant, the donor bile ducts depend on the hepatic artery, through the peribiliary plexus
- Warm ischemia during DCD injures that plexus and the duct lining
- Weeks to months later: non-anastomotic strictures, sometimes needing retransplant

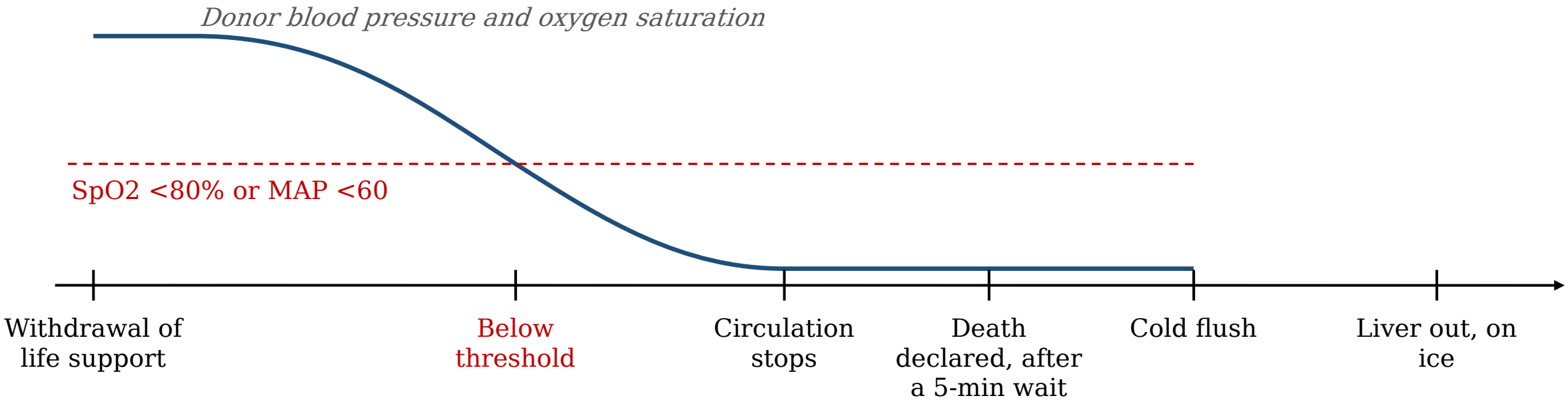
Seehofer D, Eurich D, Veltzke-Schlieker W, Neuhaus P. Am J Transplant 2013;13(2):253-65 [PMID: 23331505]

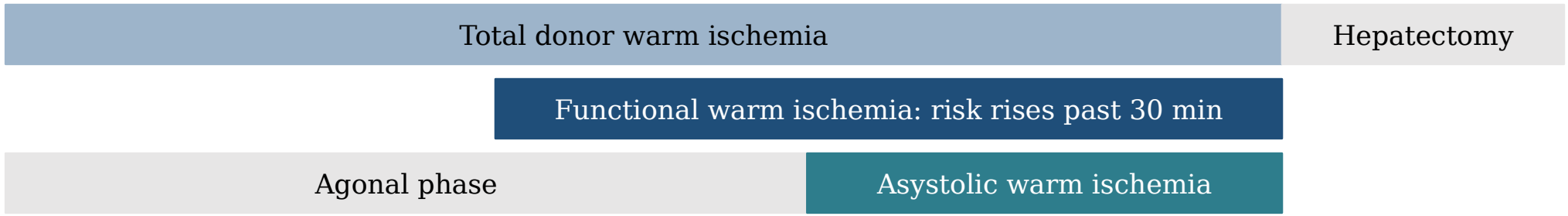
When does the warm ischemia clock start?



Definitions from Kalisvaart M, Croome KP, Hernandez-Alejandro R, et al. ILTS consensus working group. Transplantation 2021;105(6):1156-64 [PMID: 34048418]

When does the warm ischemia clock start?

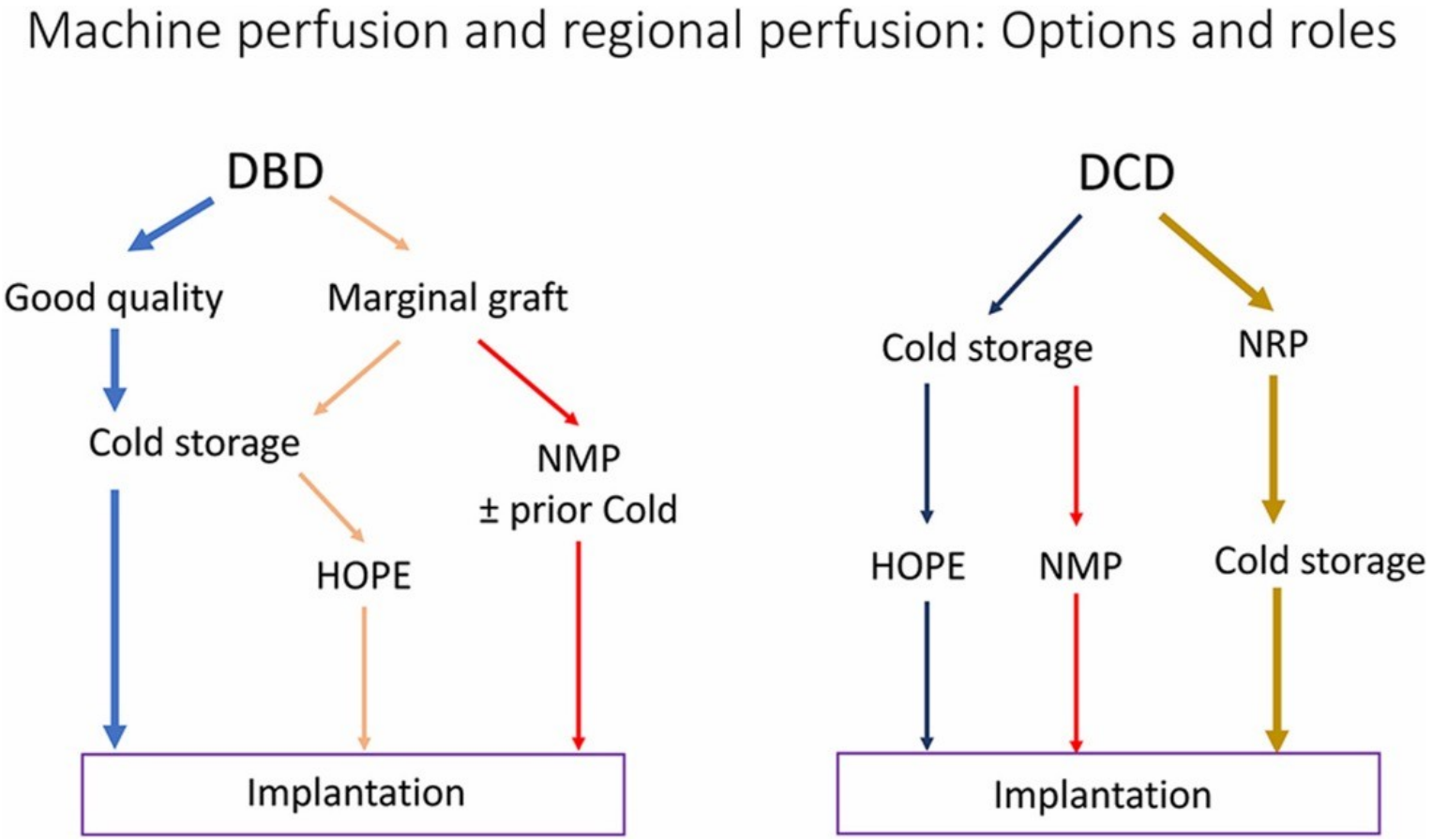




With NRP, warm oxygenated blood reaches the liver again soon after death is declared

Definitions from Kalisvaart M, Croome KP, Hernandez-Alejandro R, et al. ILTS consensus working group. Transplantation 2021;105(6):1156-64 [PMID: 34048418]

What does a perfusion machine do?

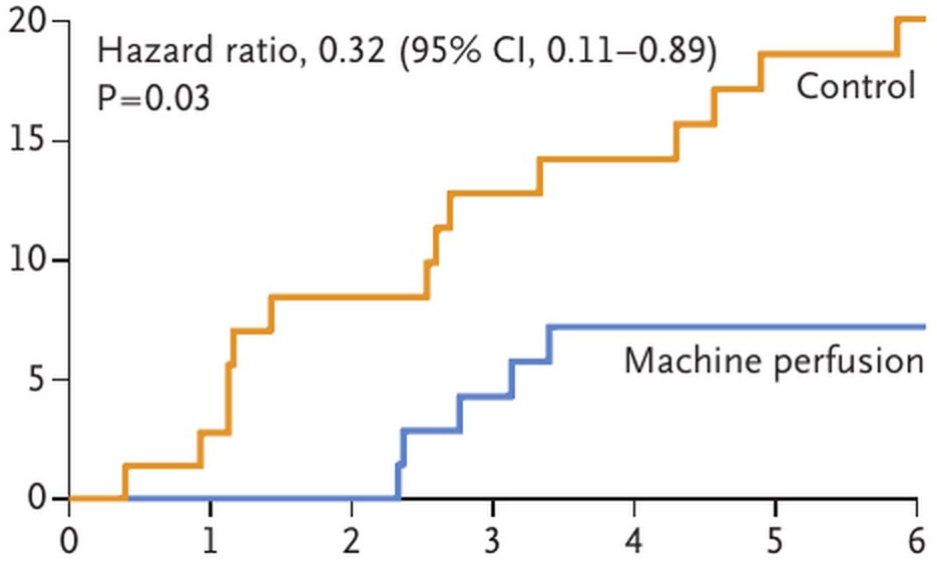


- HOPE keeps the liver cold but gives it oxygen, usually at the transplant center just before the implant
- NMP keeps it warm on blood; the liver makes bile and can be tested before anyone commits to it
- NRP restarts oxygenated blood flow to the donor's abdominal organs after death is declared

J Clin Exp Hepatol 2024;14(2):101309 [PMID: 38274508]

What did the trials show?

Trial	Machine	Main result
Nasralla, Nature 2018	NMP	Peak AST and discard 50% lower
PROTECT, JAMA Surg 2022	NMP (OCS)	Early allograft dysfunction 18% vs 31% Ischemic biliary 2.6% vs 9.9%
DHOPE-DCD, NEJM 2021	DHOPE	Symptomatic non-anastomotic strictures 6% vs 18% at 6 months, RR 0.36



DHOPE-DCD: symptomatic non-anastomotic strictures (%), by month

The 2025 AASLD/AST and 2024 EASL guidelines both recommend machine perfusion for DCD and extended-criteria livers.

Nasralla D et al. Nature 2018 [PMID: 29670285]. Markmann JF et al. JAMA Surg 2022 [PMID: 34985503]. van Rijn R et al. N Engl J Med 2021 [PMID: 33626248]. Te HS et al. 2026, statements 6-10 [PMID: 40844852]. EASL 2024, p.1062

Does NRP help the liver, and is it allowed?

17 US centers, 2017-2023	NRP (106) vs super-rapid (136)
Ischemic cholangiopathy	
Early allograft dysfunction	
Hospital stay, median	
Primary nonfunction	
Patient and graft survival	

Brubaker AL et al. JAMA Surg 2024;159(6):677-85 [PMID: 38568597]. OPTN/HRSA NRP safety notice, November 21, 2025. HRSA, NRP policy issue page (accessed 10/2026)

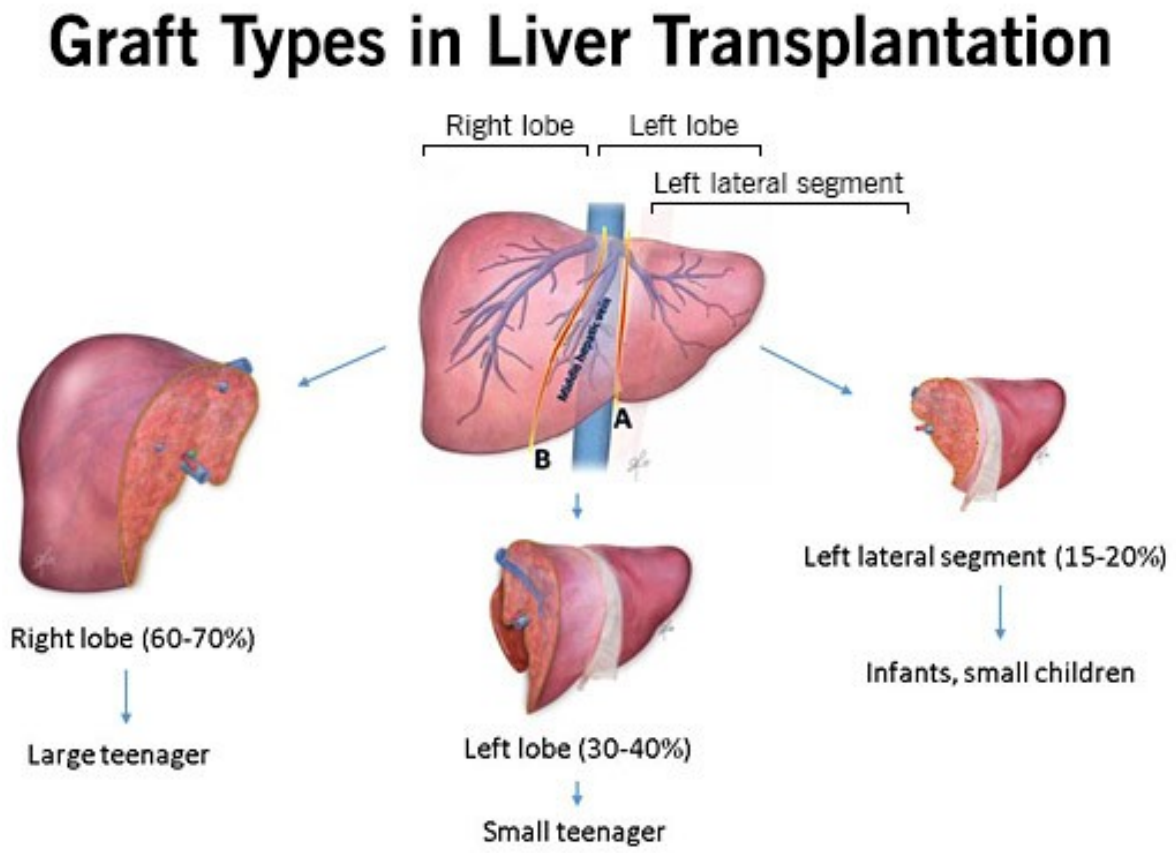
Does NRP help the liver, and is it allowed?

17 US centers, 2017-2023	NRP (106) vs super-rapid (136)
Ischemic cholangiopathy	Lower with NRP
Early allograft dysfunction	Lower with NRP
Hospital stay, median	7 vs 10 days
Primary nonfunction	0 vs 2 grafts
Patient and graft survival	Similar

- No federal or state law names NRP; the argument is over the dead donor rule
- November 2025: OPTN/HRSA safety notice on cutting and venting the vessels to the brain
- An OPTN workgroup has been drafting national NRP policy since October 2025

Brubaker AL et al. JAMA Surg 2024;159(6):677-85 [PMID: 38568597]. OPTN/HRSA NRP safety notice, November 21, 2025. HRSA, NRP policy issue page (accessed 10/2026)

What about living donors?



- 4.5% of US adult liver transplants in 2024
- Aim for a graft-to-recipient weight ratio of 0.8% or more
- Caution above MELD 30
- Older recipient age alone doesn't exclude

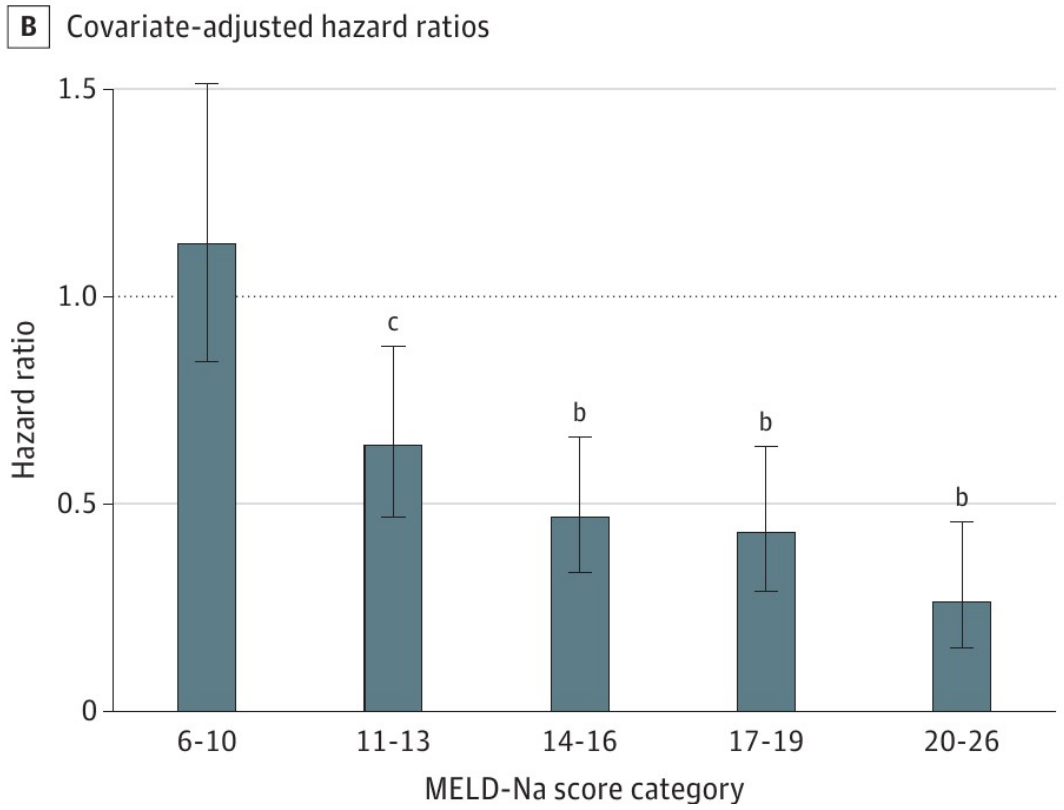
Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234], statements 50-53. Kwong AJ et al. OPTN/SRTR 2024 Annual Data Report: Liver. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

Is there benefit below MELD 15 with a living donor?

JAMA Surg. 2022 Oct 1;157(10):926-932. doi: 10.1001/jamasurg.2022.3327.

Survival Benefit of Living–Donor Liver Transplant

Whitney E Jackson [1 2], John S Malamon [2 3], Bruce Kaplan [1 2 3], Jessica L Saben [2 3], Jesse D Schold [4], James J Pomposelli [2 3], Elizabeth A Pomfret [2 3]

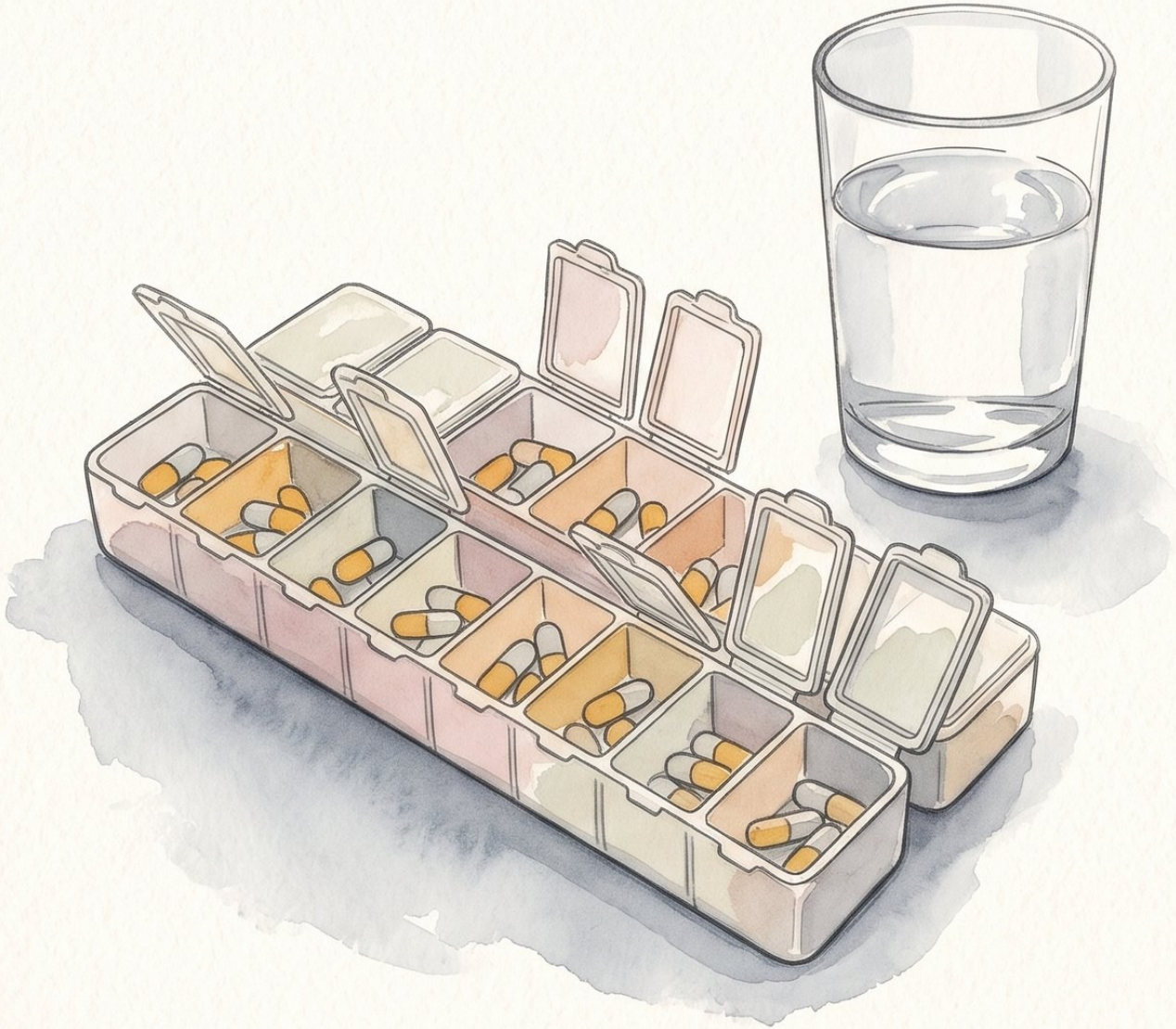


Adjusted, compared with staying on the list:

- MELD-Na 11-13: aHR 0.64 (0.47-0.88)
- MELD-Na 14-16: aHR 0.47 (0.34-0.66)
- Modeled gain of 13-17 life-years

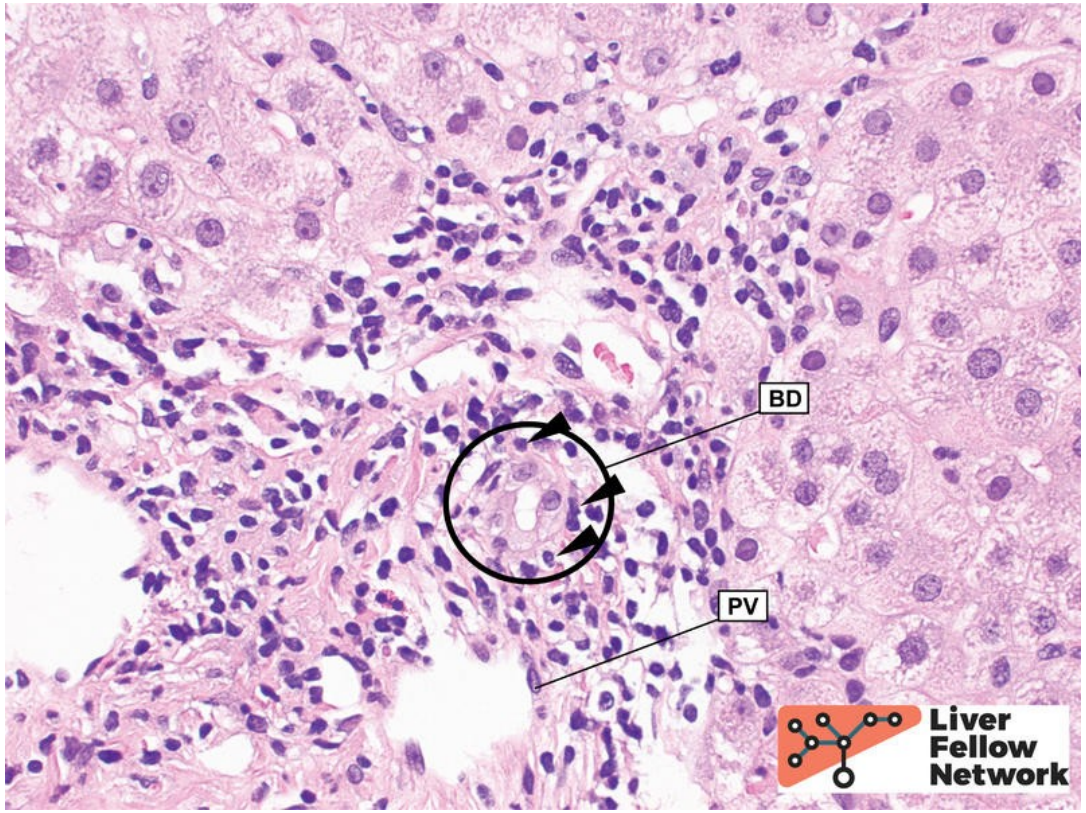
Jackson WE, Malamon JS, Kaplan B, Saben JL, Schold JD, Pomposelli JJ, Pomfret EA. JAMA Surg 2022;157(10):926-932 [PMID: 35921119]

Your patient had a transplant



Abnormal liver tests after transplant: what next?

- New or rising liver tests: call transplant hepatology the same day
- Look at the timing, the pattern, new drugs and old labs
- Doppler ultrasound and a correctly timed tacrolimus trough
- Think vascular, biliary, infection, drugs and rejection
- The transplant team decides on a biopsy



Acute cellular rejection

Te HS et al. AASLD/AST graft-related complications guideline. Liver Transpl 2026;32(3):444-90 [PMID: 40844852], statement 34. Histology: Liver Fellow Network

MRCP or ERCP for a suspected stricture?



Situation	Next step
Suspected stricture	**MRCP first (strong)**
High suspicion, or intervention likely	ERCP directly (weak)
Normal MRCP, picture still fits	ERCP (weak)
Progressive ischemic cholangiopathy	Consider retransplant

Te HS et al. AASLD/AST graft-related complications guideline. Liver Transpl 2026;32(3):444-90 [PMID: 40844852], statements 15, 16, 24. Kohli DR et al. ASGE guideline. Gastrointest Endosc 2023;97(4):607-14 [PMID: 36797162]

Which drugs change tacrolimus levels?

Raise levels (toxicity)	Lower levels (rejection)
Fluconazole, voriconazole, posaconazole	Rifampin
Clarithromycin, erythromycin	Phenytoin, carbamazepine
Diltiazem, verapamil	Phenobarbital
Nirmatrelvir/ritonavir (Paxlovid)	St John's wort
Ritonavir- or cobicistat-boosted HIV drugs	
Grapefruit, cannabidiol (CBD)	

Starting or stopping a drug, or the patient can't take pills: call transplant pharmacy or hepatology for a plan.

Tacrolimus (Prograf) prescribing information. Sharma P et al. AASLD/AST non-graft complications guideline. Liver Transpl 2026;32(8):1157-93 [PMID: 41405235]

What does the primary care doctor need to do?

	What the 2025 AASLD/AST guideline says
Vaccines	ACIP schedule for immunocompromised adults; no live vaccines
Skin	Skin exam every year, plus age-appropriate cancer screening
Alcohol	None, whatever the original liver disease
Type 2 diabetes	GLP-1 RA or SGLT2i first if high CV, heart failure or CKD risk
Lipids	Hydrophilic statin first
Elective surgery	At least 3 months after transplant, with stable graft function

Sharma P et al. AASLD/AST non-graft complications guideline. Liver Transpl 2026;32(8):1157-93 [PMID: 41405235], statements 9, 19-20, 44-45, 59, 62, 83

Take-home points

- Refer after any decompensating event. Without one, MELD 15 is still reasonable, and a low MELD alone shouldn't stop a referral
- Acute liver failure or ACLF: call the same day
- Recent drinking calls for assessment and treatment, not an automatic no
- Colorectal metastases and selected cholangiocarcinoma are worth a call
- After transplant, abnormal liver tests mean a call to transplant hepatology the same day
- Check every new drug against tacrolimus



AASLD/AST guidelines: candidate evaluation [PMID: 41405234], graft complications [PMID: 40844852], non-graft complications [PMID: 41405235]

Questions?

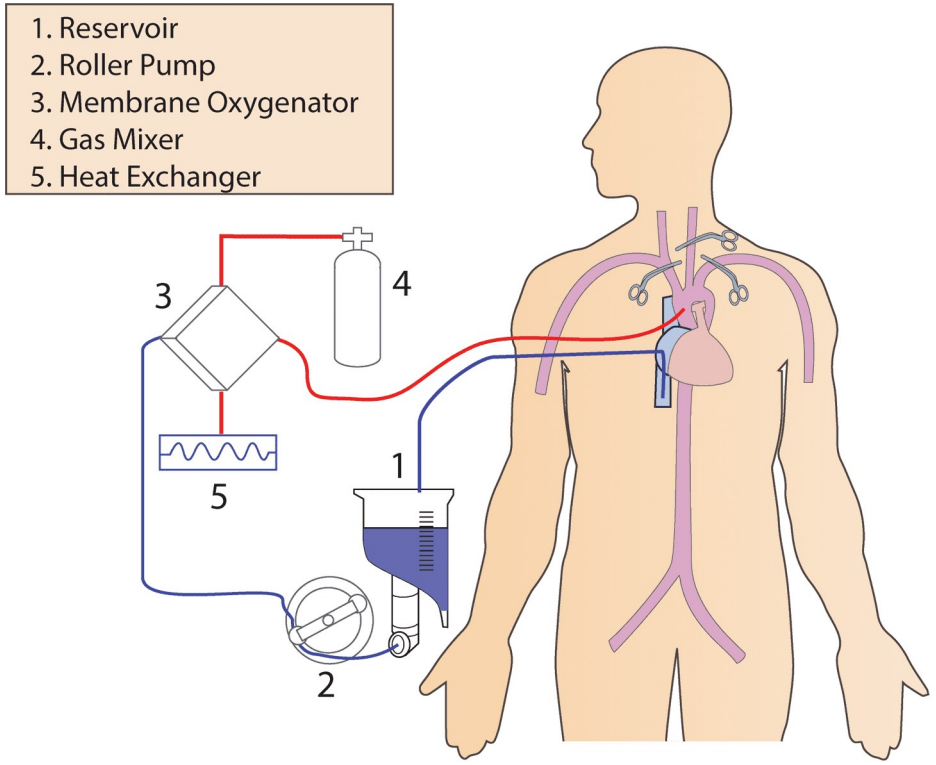


Who can get a transplant for intrahepatic CCA?

	With cirrhosis	Without cirrhosis
Tumor	Solitary, unresectable, <3 cm	Unresectable
Biopsy	Required	Required
Before listing	6 months stable after treatment	≥6 months stable on systemic therapy
Outside the liver	No disease	No disease
Setting	Exception points by review (MMaT-3)	Research protocols only

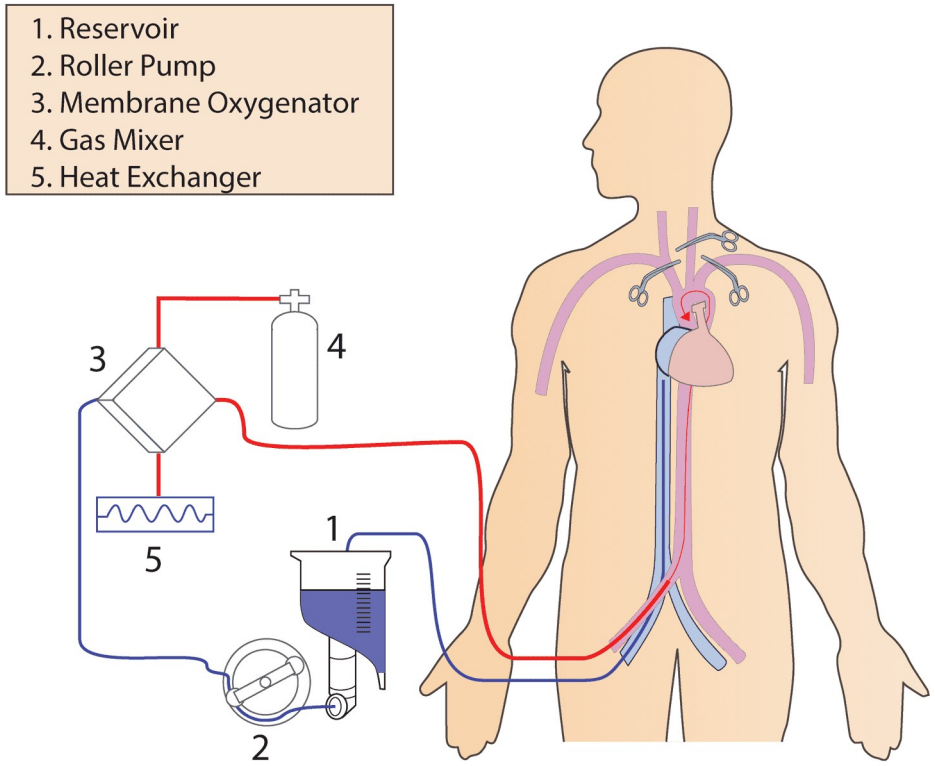
Kodali S, Kulik L, D'Alessio A, et al. ILTS-ILCA consensus. Liver Transpl 2025;31(6):815-31 [PMID: 40014003]. Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234], statement 11. OPTN transplant oncology review guidance

How do TA-NRP and A-NRP differ?



Thoraco-abdominal normothermic regional perfusion

TA-NRP: the chest is opened and the arteries to the head are clamped, cut and vented. The heart beats again, so heart, lungs and abdominal organs can all be used.



Abdominal normothermic regional perfusion

A-NRP: the aorta is blocked above the celiac and cut above the block. Only the abdominal organs get blood; the heart doesn't restart.

Figure: Alamouti-Fard E, Garg P, Wadiwala IJ, et al. Cureus 2022;14(6):e26437, CC BY 4.0. Vessel steps: OPTN/HRSA NRP safety notice, November 21, 2025

Appendix: sources (1)

- AASLD/AST candidate evaluation guideline. Hepatology 2026 [PMID: 41405234]
- AASLD/AST graft-related complications guideline. Liver Transpl 2026 [PMID: 40844852]
- AASLD/AST non-graft complications guideline. Liver Transpl 2026 [PMID: 41405235]
- EASL CPG on liver transplantation. J Hepatol 2024 [PMID: 39487043]
- ACG alcohol-associated liver disease guideline. Am J Gastroenterol 2024 [PMID: 38174913]
- ACG acute liver failure guideline. Am J Gastroenterol 2023 [PMID: 37377263]
- AASLD ACLF and critically ill cirrhosis guidance. Hepatology 2024 [PMID: 37939273]
- Lai JC et al. Frailty and sarcopenia guidance. Hepatology 2021 [PMID: 34233031]
- Kodali S et al. ILTS-ILCA iCCA consensus. Liver Transpl 2025 [PMID: 40014003]
- OPTN Policy 9 (effective 10/1/2026) and transplant oncology review guidance
- OPTN/SRTR Annual Data Reports, Liver, 2020-2024 [PMIDs: 35266621, 37132348, 38431359, 39947804, 42276697]

Appendix: sources (2)

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- Dukewich M et al. Early LT for AH, 5-year outcomes. Am J Gastroenterol 2025 [PMID: 38994850]

Appendix: sources (3)

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- Gustot T et al. CHANCE. J Hepatol 2026 [PMID: 42822699]
- Adam R et al. TransMet. Lancet 2024 [PMID: 39306468]
- Chávez-Villa M et al. LT for colorectal metastases in the US. Liver Transpl 2026 [PMID: 41086433]
- Giorgakis E et al. DCD and machine perfusion in the US. Transpl Int 2026 [PMID: 42294094]
- Seehofer D et al. Non-anastomotic strictures. Am J Transplant 2013 [PMID: 23331505]
- Kalisvaart M et al. ILTS donor warm ischemia time. Transplantation 2021 [PMID: 34048418]
- Machine and regional perfusion options. J Clin Exp Hepatol 2024 [PMID: 38274508]
- Nasralla D et al. Nature 2018 [PMID: 29670285]; Markmann JF et al. PROTECT. JAMA Surg 2022 [PMID: 34985503]
- van Rijn R et al. DHOPE-DCD. N Engl J Med 2021 [PMID: 33626248]
- Kohli DR et al. ASGE post-LT biliary strictures. Gastrointest Endosc 2023 [PMID: 36797162]
- Tacrolimus (Prograf) prescribing information
- The New York Times, March 5, 2017 (Starzl obituary)

Appendix: sources (4)

- Brubaker AL et al. NRP vs super-rapid recovery. JAMA Surg 2024 [PMID: 38568597]
- Alamouti-Fard E et al. NRP in DCD heart transplantation. Cureus 2022;14(6):e26437
- OPTN/HRSA NRP safety notice, November 21, 2025, and HRSA NRP policy issue page